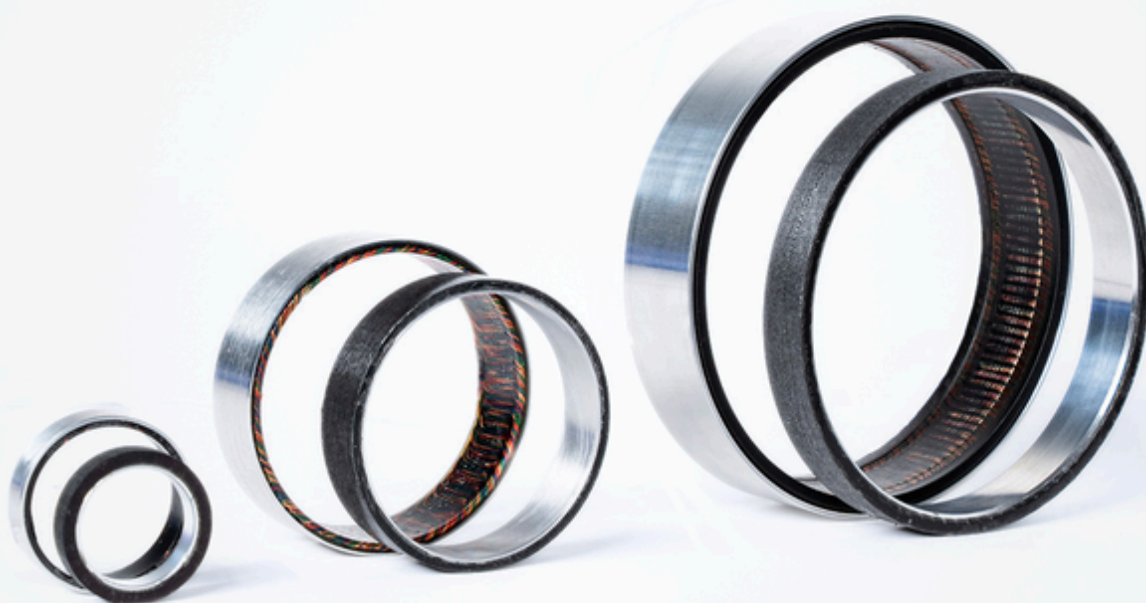


Product Catalog

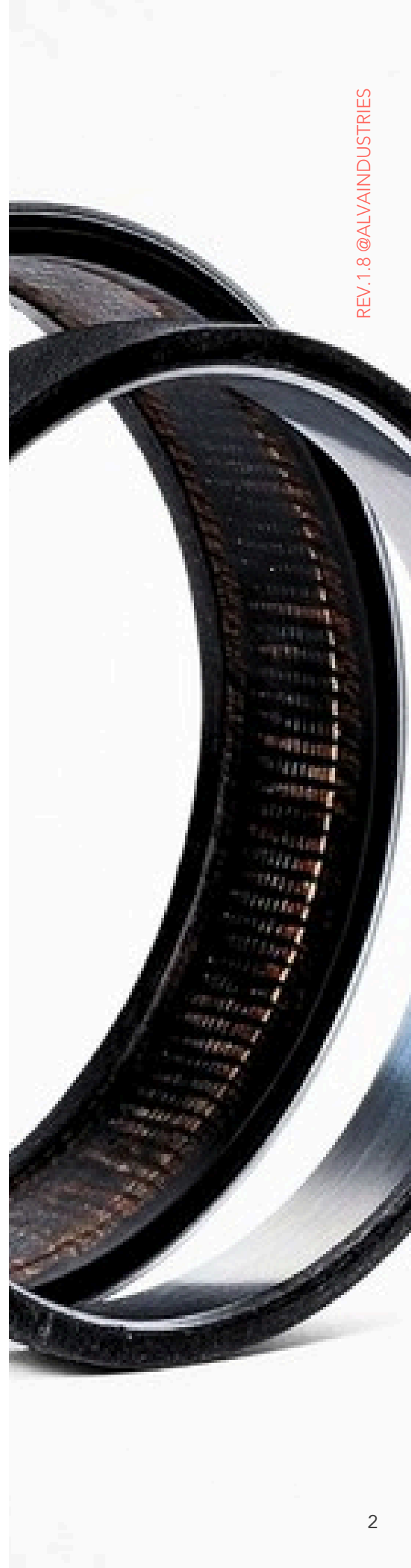
SlimTorq™ Motors

Precision Torque Motors



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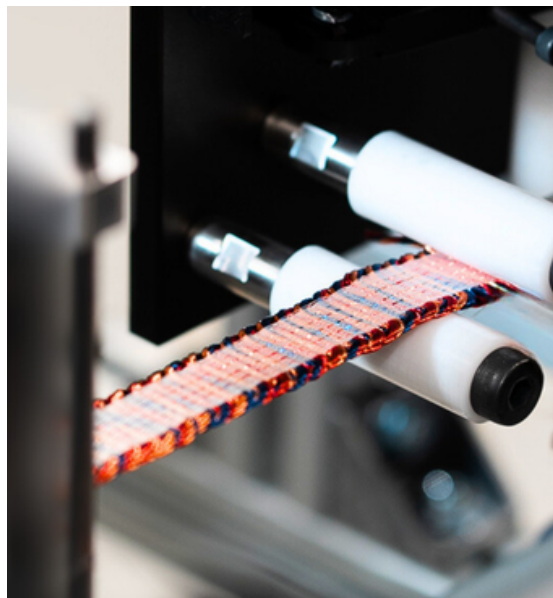


About Us

Purpose & mission

In Alva, we envision a world where everyone enjoys a high standard of living, made possible through universal access to energy and human-centered technology.

Our purpose is to create affordable, energy-efficient motors and generators that improve lives and protect the planet.



Safety & reliability by design

We direct our passion for electric solutions into every product & system that leaves our production facility. With complete control of the entire value chain, we monitor every component all the way from design to finished product, with full traceability along the way. We design our motors with safety & reliability as our leading design criteria. This way, you can focus on what really matters, & rest assured that your Alva system will perform in any & all conditions.

Our Origin

Alva was founded in 2017 in Trondheim, Norway, starting as a passion project in our founder Jørgen Selnes' garage. What began as an idea to rethink electric motor technology has grown into a dedicated company focused on pushing the boundaries of what electric motors can achieve. With a strong foundation of innovation & collaboration, we continue to develop groundbreaking solutions that meet the evolving needs of various industries.

Industries & Applications

Powering the Future

At Alva, we're pioneering electrification with innovative motor technology, driving sustainable transformation within defense, medical & robotics. With FiberPrinting™ technology, we engineer the reliability & efficiency to propel industries forward.



High Precision Gimbals

With Alva's FiberPrinted™ slotless motors the Size, Weight & Power of Gimbals can be drastically improved through high precision & high gravimetric torque density.

Medical Devices

The medical industry is advancing by utilizing robot solutions to rehabilitate & heal patients faster & in a less invasive manner. Alva's cogging free motors improve the precision of medical devices like surgical robotics & exoskeletons.

Marine Propulsion

FiberPrinted™ motors offer high efficiency, reliability, & a market-leading ID/OD ratio for marine propulsion systems, making them ideal for rim-drive thrusters in cleaning ROVs & long-range AUVs & UUVs, where minimal drag & low speed-dependent losses are critical.

Defense & SatCom

Alva motors designed for defense, satellite communication, reaction wheels, laser communication, & high-speed auxiliary power units, will offer precision, low cogging, lightweight designs, & low power usage for critical applications in dynamic environments.

Metrology

The physical world has become more digital than ever before. Almost all industries rely on measurement data, including manufacturing, construction, medical, & aerospace. To maximize efficiency & throughput within these industries, measurements are increasingly automated, & Alva's high-precision motors support this.

Light Robotics & Humanoids

Alva Industries' FiberPrinted™ motors offer high torque density & large through-hole, making them ideal for compact & strong actuators with gearboxes integrated in the center of the motor. SlimTorq motors are cogging free which makes them highly transparent & easy to include in a Sim2Real workflows.

FiberPrinting™

A Patented Technology

Alva's next-gen electric motors are manufactured by our patented FiberPrinting™ technology. This groundbreaking & 6 times patented process enables us to rapidly design & manufacture ironless & slotless windings with high copper fill factor for any electric motor or generator, setting a new standard in precision, torque-density & adaptability.



The FiberPrinting™ Process

Alva's FiberPrinting™ revolutionizes stator windings for rotary, linear, and arc motors. With a 60% copper fill factor, it boosts efficiency, torque, and power. The process enables custom stators for precision & performance applications across industries like marine, medical, and aerospace.

- High Copper Fill: Maximizes power, reduces resistance.
- Non-Skewed Windings: Perfect magnetic alignment.

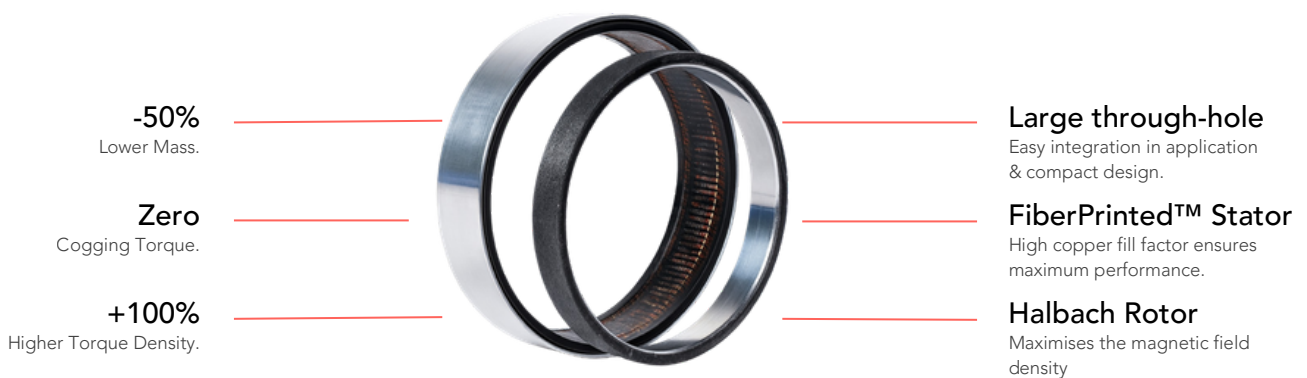
The Advantage

Alva's next-generation electric motors are powered by FiberPrinting™, our six-time patented technology. This never before seen method allows us to design & produce ironless, slotless windings with unmatched precision & torque density. With a high copper fill factor, custom shapes, & non-skewed windings, FiberPrinting™ pushes the limits of electric motor performance - delivering higher power, efficiency, & reliability for any application.

SlimTorq™ Motors

Precision & Performance Unified

SlimTorq™ is a slotless thin-section frameless motor series for high-precision applications. The SlimTorq™ motor series has zero cogging, high torque density, no saturation & large through-hole.



Performance

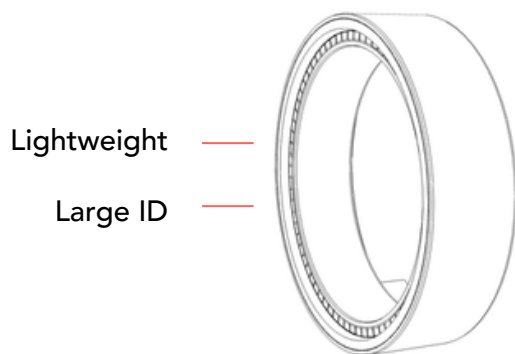
Alva's proprietary manufacturing technology allows for maximised copper fill factor & non-skewed stator windings, which result in a motor-constant & torque-density rivalling even the best slotted motors.

Light & Compact

The removal of the iron-teeth in the SlimTorq™ motors result in a stator & rotor that's astonishingly thin & light. The motors feature a large through-hole that allows for design of incredible compact actuators.

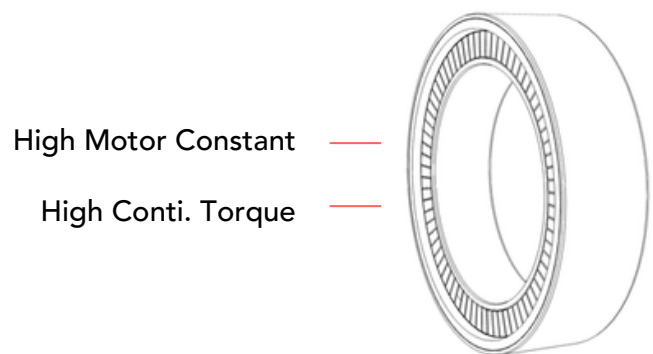
Precision

SlimTorq™ torque motors are slotless & therefore has virtually zero cogging torque & minor torque ripple. Making them the perfect choice for high-precision applications that also require high-performance.



SlimTorq™ Lite

The Lite version is designed & optimised to be as light as possible while sustaining a high torque-density. It's a good choice for weight sensitive applications. It allows has the added value of a large through-hole to simply mechanical integration.



SlimTorq™ Max

The Max version is designed & optimised for maximum performance i.e. as high as possible torque and motor-constant that the space envelope allows. It's a good choice for applications that require the maximum performance in combination of compactness.

Custom Sizes

Custom Size For Any Application

Alva offers custom SlimTorq™ motor sizes when a standard size doesn't fit your application. Contact us at sales@alvaindustries.com if you are interested or design your own custom motor with our web-based design suite [TorqStudio](#).

Outer Diameter

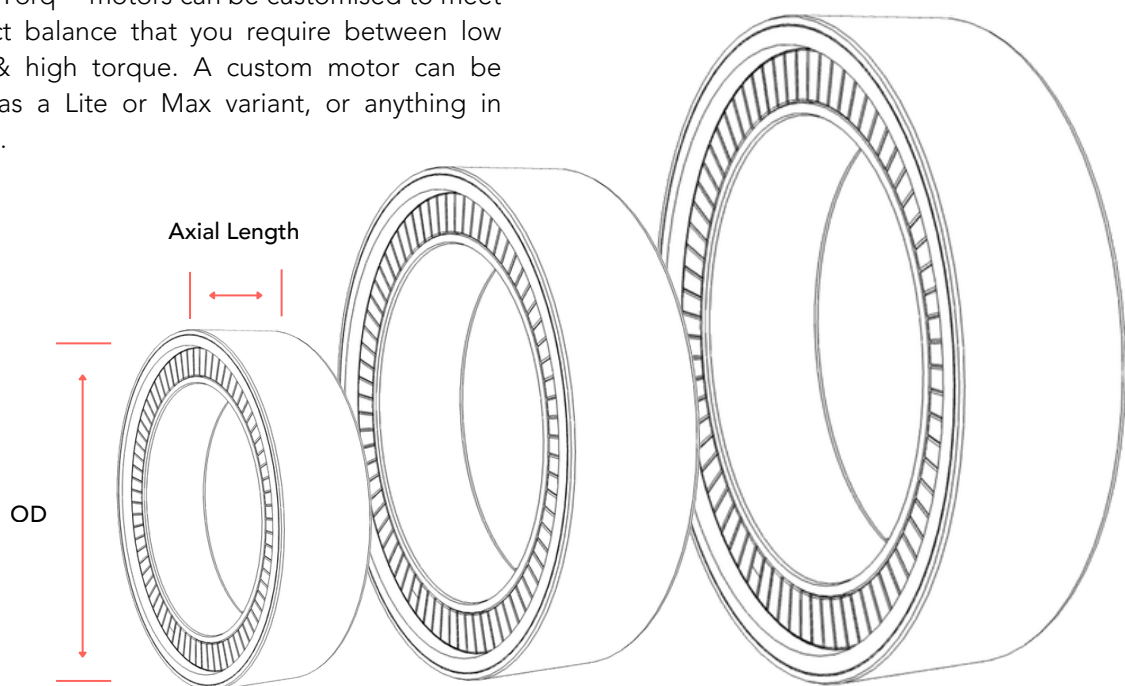
Due to the flexibility of FiberPrinting™, the outer diameter can be adjusted according to your exact needs with no fixed increment limitation.

Axial Length

The axial length can be adjusted to fit your mechanical assembly perfectly with no fixed increment limitation.

Lite & Max

The SlimTorq™ motors can be customised to meet the exact balance that you require between low weight & high torque. A custom motor can be offered as a Lite or Max variant, or anything in between.



TorqStudio: Design your own motor

TorqStudio allows you to design & analyse your own motor with a fast & easy to use web-based interface. Access it today to explore the possibilities at [TorqStudio](#).

Contact us at sales@alvaindustries.com to access information about timeline, MOQ & NRE of custom motors.



SlimTorq™ Motors

One Size For Every Need

The FiberPrinting™ technology allows for production of motors in almost any size. The SlimTorq™ product portfolio includes motor sizes from 25 mm up to 190 mm.

	Motor	Motor constant [Nm/√W]	Continuous torque [Nm]	Rated voltage [V]	Stator OD [mm]	Axial length [mm]	Rotor ID [mm]	Total mass [g]
SlimTorq™ Lite	75-20-L	0.132	0.654	72	75	20	61	126.0
	85-24-L	0.188	1.06	72	85	24	69	200.0
	105-17-L	0.214	1.14	72	105	17	89	177.0
	130-27-L	0.296	2.03	72	130	27	112	337.0
	160-17-L	0.377	2.34	72	160	17	140	318.0
	190-35-L	0.8410	8.72	72	190	35	157	1089.0
SlimTorq™ Max	25-9-M	0.00948	0.0232	72	25	9	12.3	12.9
	39-10-M	0.0194	0.0577	72	39	10	25	27.5
	51-12-M	0.041	0.144	72	51	12	37	46.2
	75-20-M	0.179	0.905	72	75	20	53	209.0
	85-24-M	0.252	1.44	72	85	24	64	291.0
	105-17-M	0.259	1.39	72	105	17	84	250.0
	130-27-M	0.507	3.51	72	130	27	107	475.0
	160-17-M	0.469	2.94	72	160	17	135	426.0
	190-35-M	1.082	11.29	72	190	35	155	1352.0

Access the latest CAD-files for all motors on our [website](#).

SlimTorq™ Motors

Configured To Your Needs

Check out the [Product Selector](#) & [TorqStudio](#) to simplify the configuration process.

Winding Options

Number of series turns

2-8 - The stator can be configured with a variant of series connections. Higher number of series turns result in higher torque/voltage constants & lower no-load speed.

Star & Delta winding connection

Y - Star connection. The three phases are connected as a star which result in higher torque/voltage constants & lower no-load speed.

D - Delta connection. The phases are connected in a triangle which result in lower torque/voltage constants & higher no-load speed.

Terminal direction options

A - Axial
R - Radial



Cable Options

Cable gauge

1-99 - All motor sizes comes with a default gauge. Cable gauge can be customised on demand.

Cable type

A - UL rated cable with PTFE wire insulation.

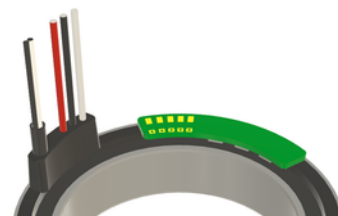
Sensor Options

Temperature sensor options

A - 10K NTC with a B25/100 value of $3492 \pm 1\% K$

Position sensor option

A - 3-phase hall-effect sensor with open-drain output, built-in $1.2 k\Omega$ pull-up resistor, & 5 VDC input voltage.



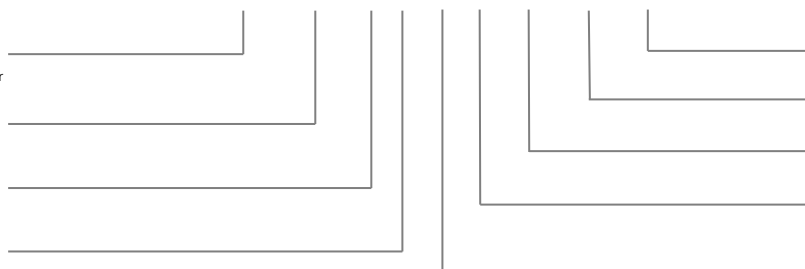
STM-130-27-L-4Y-A-18A-A0-001

Motor Series
STM - SlimTorq™ Motor

Diameter
Stator OD (mm)

Axial Length
Stator Length (mm)

Motor Variant
L - Lite
M - Max



Unique Identifier

Sensor Options
Temperature + Position (A-Z) + (A-Z) (0 = None)

Cable Gauge & Type
AWG(1-99) + Type(A-Z)

Terminal Option
A - Axial Direction
R - Radial Direction

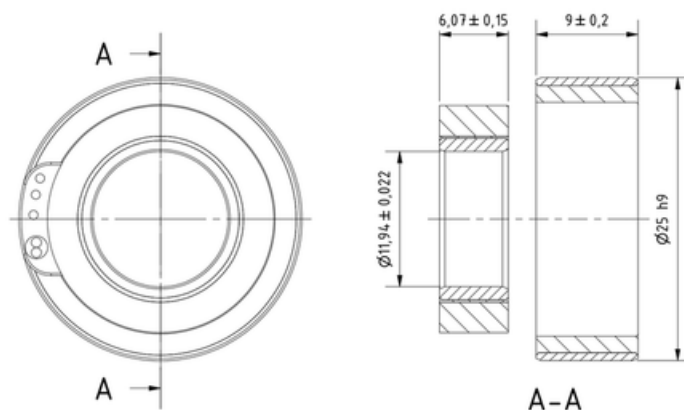
Winding Option
Series Turns + Star(Y)/Delta(D)

Motor	Variant options	Series turns options	Connection options	Terminal direction options	Standard cable	Temp sensor options	Position sensor options
25-9	M - Max	3, 6	Y - Star D - Delta	A - Axial R - Radial	26A	A - 10K NTC	A - Hall-effect
39-10	M - Max	2, 4	Y - Star D - Delta	A - Axial R - Radial	26A	None	A - Hall-effect
51-12	M - Max	2, 4, 8	Y - Star D - Delta	A - Axial R - Radial	24A	A - 10K NTC	A - Hall-effect
75-20	L - Lite M - Max	2, 4, 8	Y - Star D - Delta	A - Axial R - Radial	24A	A - 10K NTC	A - Hall-effect
85-24	L - Lite M - Max	2, 4, 8	Y - Star D - Delta	A - Axial R - Radial	24A	A - 10K NTC	A - Hall-effect
105-17	L - Lite M - Max	2, 4, 8	Y - Star D - Delta	A - Axial R - Radial	22A	A - 10K NTC	A - Hall-effect
130-27	L - Lite M - Max	2, 4, 8	Y - Star D - Delta	A - Axial R - Radial	18A	A - 10K NTC	A - Hall-effect
160-17	L - Lite M - Max	2, 4, 8	Y - Star D - Delta	A - Axial R - Radial	18A	A - 10K NTC	A - Hall-effect
190-35	L - Lite M - Max	2, 4, 8	Y - Star D - Delta	A - Axial R - Radial	16A	A - 10K NTC	A - Hall-effect

SlimTorq™ 25-9

PROPERTY	UNIT	STM-25-9-M			
Motor type	-	Max			
Winding type	-	3D	3Y	6D	6Y
PERFORMANCE					
Motor constant ¹	Nm/√W	0.00948			
Continuous torque ²	Nm	0.023			
Peak torque, 1 s ³	Nm	0.116			
Peak torque, 3 s ³	Nm	0.083			
Peak torque, 5 s ³	Nm	0.070			
Cogging torque	Nm	0.0			
Max continuous current ²	Arms	7.89	4.56	3.95	2.28
Peak current, 1 s ³	Arms	39.3	22.7	19.6	11.3
Peak current, 3 s ³	Arms	28.3	16.4	14.2	8.18
Peak current, 5 s ³	Arms	23.9	13.8	11.9	6.89
Max speed at max voltage ⁴	RPM	10000	10000	10000	10000
ELECTRICAL					
Torque constant ¹	Nm/Arms	0.0029	0.0051	0.0059	0.0102
Voltage constant ¹	V/kRPM	0.251	0.437	0.502	0.874
Line to line resistance ¹	Ohm	0.064	0.192	0.257	0.770
Line to line inductance ¹	μH	0.562	1.690	2.250	6.740
Electrical time constant	μs	8.76			
Spatial harmonic torque ripple	%	0.32			

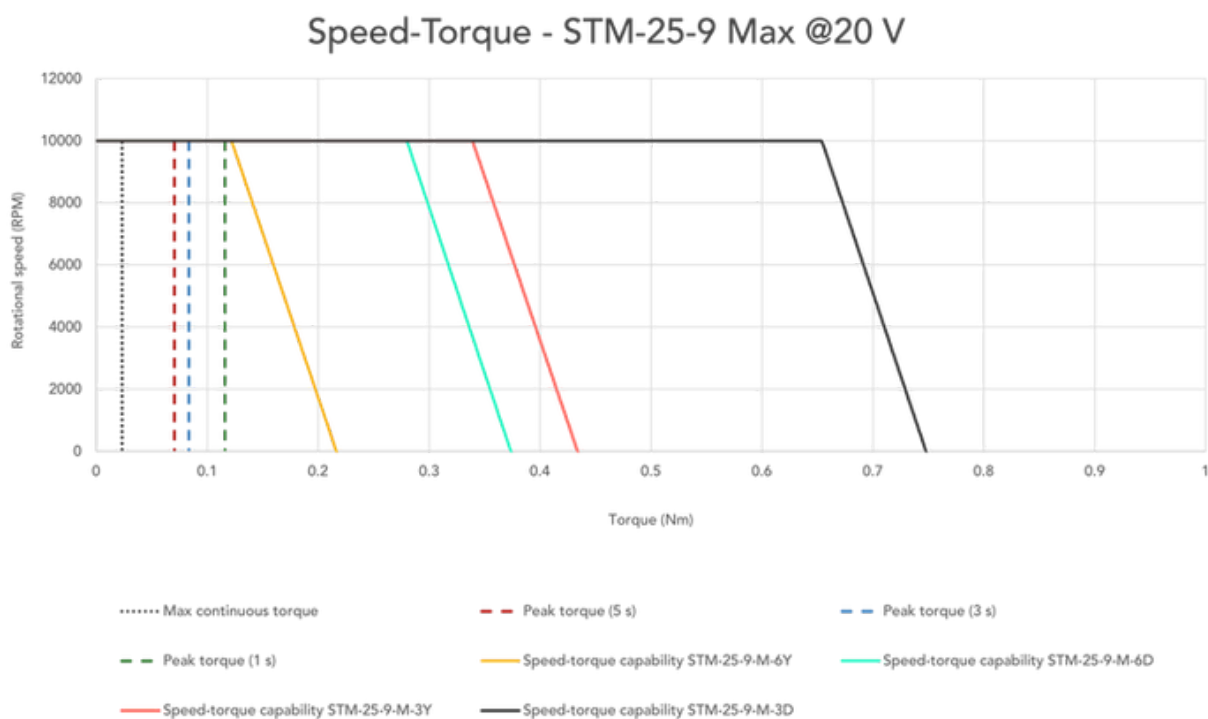
PROPERTY	UNIT	STM-25-9-M
Rated voltage	V	72
Poles	-	16
Thermal resistance ²	C/W	10.8
Max stator temperature	°C	140
Max rotor temperature	°C	85
Continuous power loss ²	W	11.2
MECHANICAL		
Stator outer diameter (OD)	mm	25
Stator OD tolerance	mm	h9
Rotor inner diameter (ID)	mm	12.3
Rotor ID tolerance	mm	+/-0.022
Stator axial length	mm	9
Stator axial length tolerance	mm	+/- 0.2
Rotor axial length	mm	6.07
Rotor axial length tolerance	mm	+/- 0.15
Rotational inertia	gcm ²	5.16
Stator mass ⁵	g	5.74
Rotor mass	g	7.09
Total mass ⁵	g	12.9
Power cable length	mm	500
Power cable size	-	AWG26
ENVIRONMENTAL		
Minimum ambient temperature	°C	-40
Maximum ambient temperature	°C	80
Ambient conditions	°C	20



¹Windings at 20 °C & w/o AC cables. ²Housing without fins, 5 m/s airflow, 20°C ambient temp (conservative cooling assumptions, contact us for a continuous torque analysis for your application).

³Winding temperature increases from 20 °C to 140 °C. ⁴Lowest of max mechanical speed & max electrical speed at rated voltage. ⁵W/o AC cables.

Disclaimer: The values & specifications provided are for reference only & may be subject to change or contain inaccuracies.



*Speed-Torque curves can be generated for your specific supply voltage,
please contact sales@alvaindustries.com if you require this.

SlimTorq™ 39-10

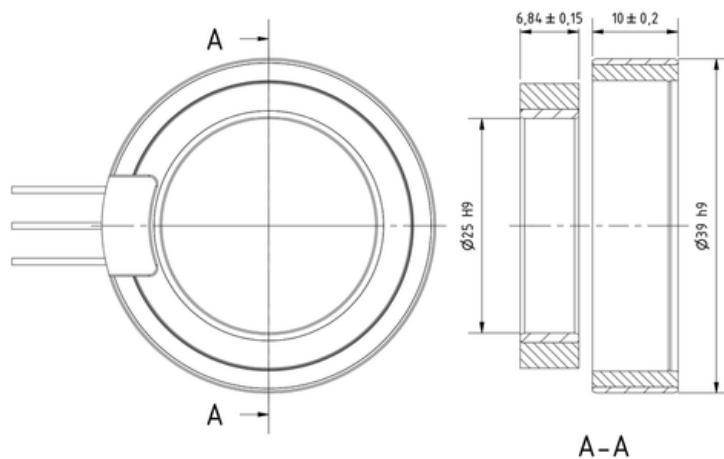
PROPERTY	UNIT	STM-39-10-M			
Motor type	-	Max			
Winding type	-	2D	2Y	4D	4Y
PERFORMANCE					
Motor constant ¹	Nm/√W	0.0194			
Continuous torque ²	Nm	0.058			
Peak torque, 1 s ³	Nm	0.31			
Peak torque, 3 s ³	Nm	0.233			
Peak torque, 5 s ³	Nm	0.199			
Cogging torque	Nm	0.0			
Max continuous current ²	Arms	8.65	4.99	4.32	2.50
Peak current, 1 s ³	Arms	46.5	26.8	23.2	13.4
Peak current, 3 s ³	Arms	34.9	20.2	17.5	10.1
Peak current, 5 s ³	Arms	29.9	17.2	14.9	8.62
Max speed at max voltage ⁴	RPM	10000	10000	10000	10000
ELECTRICAL					
Torque constant ¹	Nm/Arms	0.007	0.012	0.013	0.023
Voltage constant ¹	V/kRPM	0.568	0.991	1.140	1.980
Line to line resistance ¹	Ohm	0.079	0.237	0.316	0.948
Line to line inductance ¹	μH	0.527	1.580	2.110	6.320
Electrical time constant	μs	6.67			
Spatial harmonic torque ripple	%	0.34			

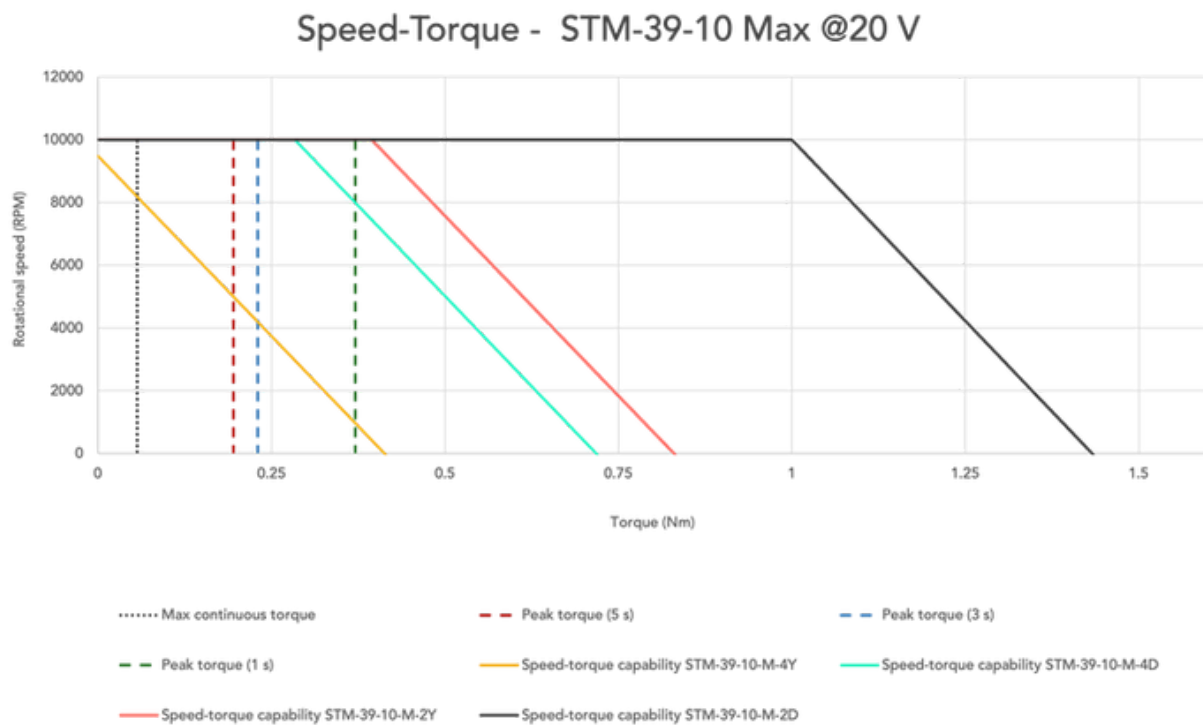
PROPERTY	UNIT	STM-39-10-M
Rated voltage	V	72
Poles	-	26
Thermal resistance ²	C/W	7.14
Max stator temperature	°C	140
Max rotor temperature	°C	85
Continuous power loss ²	W	15.3
MECHANICAL		
Stator outer diameter (OD)	mm	39
Stator OD tolerance	mm	h9
Rotor inner diameter (ID)	mm	25
Rotor ID tolerance	mm	H9
Stator axial length	mm	10
Stator axial length tolerance	mm	+/- 0.2
Rotor axial length	mm	6.84
Rotor axial length tolerance	mm	+/-0.15
Rotational inertia	gcm ²	33.9
Stator mass ⁵	g	12.6
Rotor mass	g	14.8
Total mass ⁵	g	27.5
Power cable length	mm	500
Power cable size	-	AWG26
ENVIRONMENTAL		
Minimum ambient temperature	°C	-40
Maximum ambient temperature	°C	80
Ambient conditions	°C	20

¹Windings at 20 °C & w/o AC cables. ²Housing without fins, 5 m/s airflow, 20°C ambient temp (conservative cooling assumptions, contact us for a continuous torque analysis for your application).

³Winding temperature increases from 20 °C to 140 °C. ⁴Lowest of max mechanical speed & max electrical speed at rated voltage. ⁵W/o AC cables.

Disclaimer: The values & specifications provided are for reference only & may be subject to change or contain inaccuracies.





*Speed-Torque curves can be generated for your specific supply voltage, please contact sales@alvaindustries.com if you require this.

SlimTorq™ 51-12

PROPERTY	UNIT	STM-51-12-M					
Motor type	-	Max					
Winding type	-	2D	2Y	4D	4Y	8D	8Y
PERFORMANCE							
Motor constant ¹	Nm/√W	0.041					
Continuous torque ²	Nm	0.144					
Peak torque, 1 s ³	Nm	0.934					
Peak torque, 3 s ³	Nm	0.664					
Peak torque, 5 s ³	Nm	0.561					
Cogging torque	Nm	0.0					
Max continuous current ²	Arms	11.3	6.55	5.67	3.28	2.84	1.64
Peak current, 1 s ³	Arms	73.4	42.4	36.7	21.2	18.4	10.6
Peak current, 3 s ³	Arms	52.2	30.2	26.1	15.1	13.1	7.54
Peak current, 5 s ³	Arms	44.1	25.4	22	12.7	11	6.36
Max speed at max voltage ⁴	RPM	10000	10000	10000	10000	10000	9530
ELECTRICAL							
Torque constant ¹	Nm/Arms	0.013	0.022	0.025	0.044	0.051	0.088
Voltage constant ¹	V/kRPM	1.090	1.890	2.170	3.780	4.340	7.550
Line to line resistance ¹	Ohm	0.064	0.192	0.256	0.769	1.020	3.070
Line to line inductance ¹	μH	0.745	2.24	2.98	8.95	11.90	35.80
Electrical time constant	μs	11.6					
Spatial harmonic torque ripple	%	0.75					

PROPERTY	UNIT	STM-51-12-M
Rated voltage	V	72
Poles	-	36
Thermal resistance ²	C/W	5.12
Max stator temperature	°C	140
Max rotor temperature	°C	85
Continuous power loss ²	W	21.3

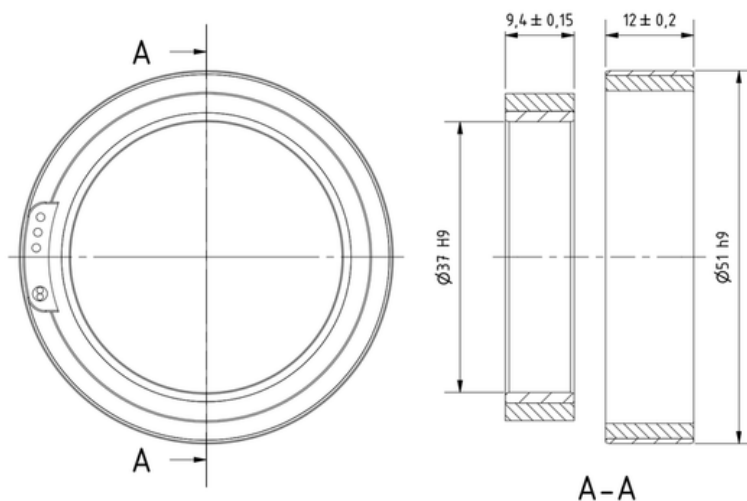
MECHANICAL		
Stator outer diameter (OD)	mm	51
Stator OD tolerance	mm	h9
Rotor inner diameter (ID)	mm	37
Rotor ID tolerance	mm	H9
Stator axial length	mm	12
Stator axial length tolerance	mm	+/- 0.2
Rotor axial length	mm	9.4
Rotor axial length tolerance	mm	+/- 0.15
Rotational inertia	gcm ²	106
Stator mass ⁵	g	22
Rotor mass	g	24.1
Total mass ⁵	g	46.2
Power cable length	mm	500
Power cable size	-	AWG24

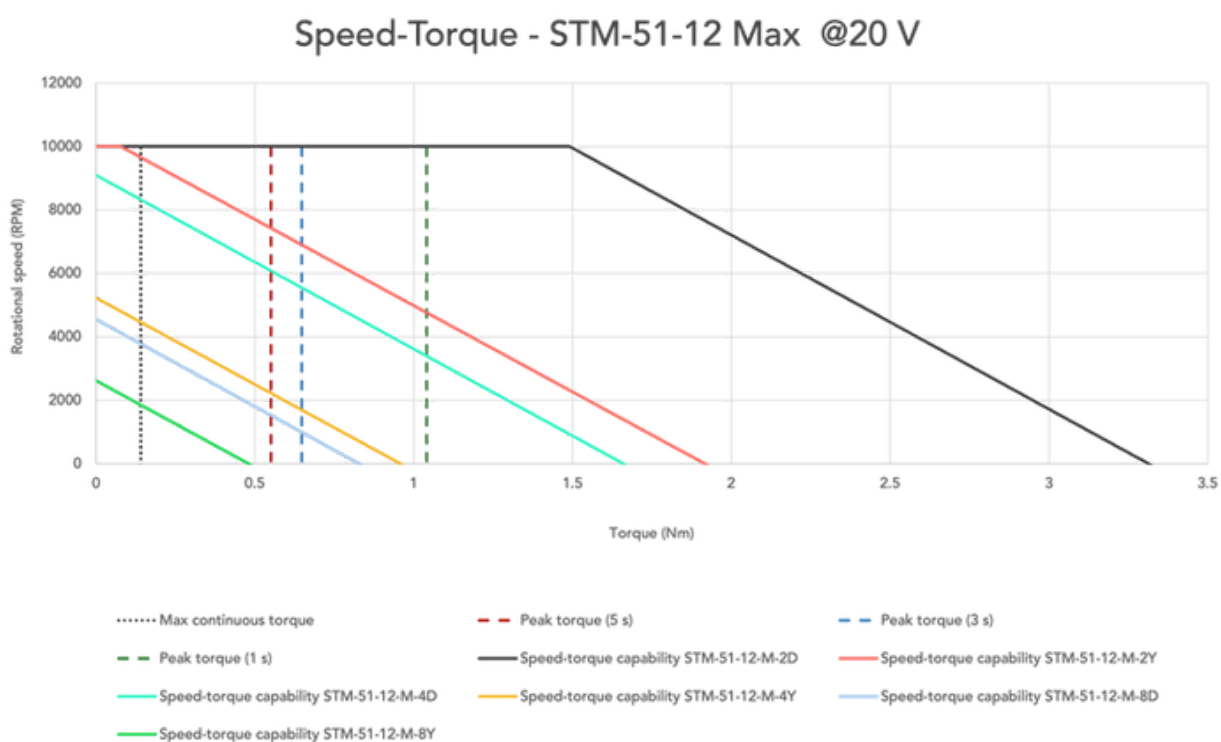
ENVIRONMENTAL		
Minimum ambient temperature	°C	-40
Maximum ambient temperature	°C	80
Ambient conditions	°C	20

¹Windings at 20 °C & w/o AC cables. ²Housing without fins, 5 m/s airflow, 20°C ambient temp (conservative cooling assumptions, contact us for a continuous torque analysis for your application).

³Winding temperature increases from 20 °C to 140 °C. ⁴Lowest of max mechanical speed & max electrical speed at rated voltage. ⁵W/o AC cables.

Disclaimer: The values & specifications provided are for reference only & may be subject to change or contain inaccuracies.





*Speed-Torque curves can be generated for your specific supply voltage,
please contact sales@alvaindustries.com if you require this.

SlimTorq™ 75-20

PROPERTY	UNIT	STM-75-20-L						STM-75-20-M					
Motor type	-	Lite						Max					
Winding type	-	2D	2Y	4D	4Y	8D	8Y	2D	2Y	4D	4Y	8D	8Y
PERFORMANCE													
Motor constant ¹	Nm/√W	0.132						0.179					
Continuous torque ²	Nm	0.654						0.905					
Peak torque, 1 s ³	Nm	5.22						7.09					
Peak torque, 3 s ³	Nm	3.56						4.83					
Peak torque, 5 s ³	Nm	2.99						4.06					
Cogging torque	Nm	0.0						0.0					
Max continuous current ²	Arms	20.7	12.00	10.40	5.98	5.18	2.99	21.10	12.20	10.60	6.09	5.28	3.05
Peak current, 1 s ³	Arms	165	95.5	82.7	47.7	41.3	23.9	165	95.5	82.7	47.7	41.3	23.9
Peak current, 3 s ³	Arms	113	65	56.3	32.5	28.2	16.3	113	65	56.3	32.5	28.2	16.3
Peak current, 5 s ³	Arms	94.7	54.7	47.4	27.3	23.7	13.7	94.7	54.7	47.4	27.3	23.7	13.7
Max speed at max voltage ⁴	RPM	8400	8400	8400	7660	6710	3830	4800	4800	4800	4800	4800	2820
ELECTRICAL													
Torque constant ¹	Nm/Arms	0.032	0.055	0.063	0.109	0.126	0.219	0.043	0.0743	0.086	0.149	0.172	0.297
Voltage constant ¹	V/kRPM	2.68	4.7	5.37	9.4	10.7	18.8	3.65	6.38	7.3	12.8	14.6	25.5
Line to line resistance ¹	Ohm	0.038	0.114	0.152	0.457	0.609	1.83	0.038	0.114	0.152	0.457	0.609	1.83
Line to line inductance ¹	μH	1.28	3.84	5.13	15.4	20.5	61.5	1.28	3.84	5.13	15.4	20.5	61.5
Electrical time constant	μs	33.7											
Spatial harmonic torque ripple	%	0.57						0.40					

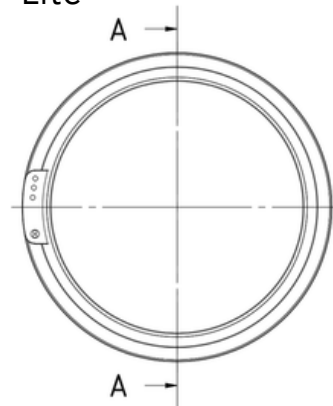
PROPERTY	UNIT	STM-75-20-L	STM-75-20-M
Rated voltage	V	72	
Poles	-	36	
Thermal resistance ²	C/W	2.59	
Max stator temperature	°C	140	
Max rotor temperature	°C	85	
Continuous power loss ²	W	42.1	42.8
MECHANICAL			
Stator outer diameter (OD)	mm	75	75
Stator OD tolerance	mm	h9	h9
Rotor inner diameter (ID)	mm	61	53
Rotor ID tolerance	mm	H9	H9
Stator axial length	mm	20	20
Stator axial length tolerance	mm	+/- 0.2	+/- 0.2
Rotor axial length	mm	15.7	15.7
Rotor axial length tolerance	mm	+/- 0.15	+/- 0.15
Rotational inertia	gcm ²	620	1350
Stator mass ⁵	g	67.1	67.1
Rotor mass	g	58.7	142
Total mass ⁵	g	126	209
Power cable length	mm	500	500
Power cable size	-	AWG 24	AWG 24
ENVIRONMENTAL			
Minimum ambient temperature	°C	-40	
Maximum ambient temperature	°C	80	
Ambient conditions	°C	20	

¹Windings at 20 °C & w/o AC cables. ²Housing without fins, 5 m/s airflow, 20°C ambient temp (conservative cooling assumptions, contact us for a continuous torque analysis for your application).

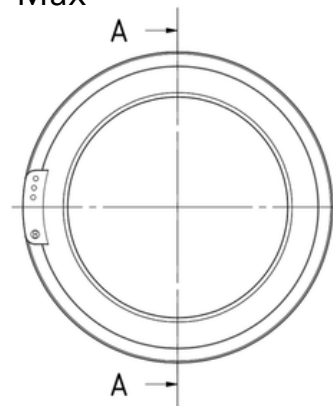
³Winding temperature increases from 20 °C to 140 °C. ⁴Lowest of max mechanical speed & max electrical speed at rated voltage. ⁵W/o AC cables.

Disclaimer: The values & specifications provided are for reference only & may be subject to change or contain inaccuracies.

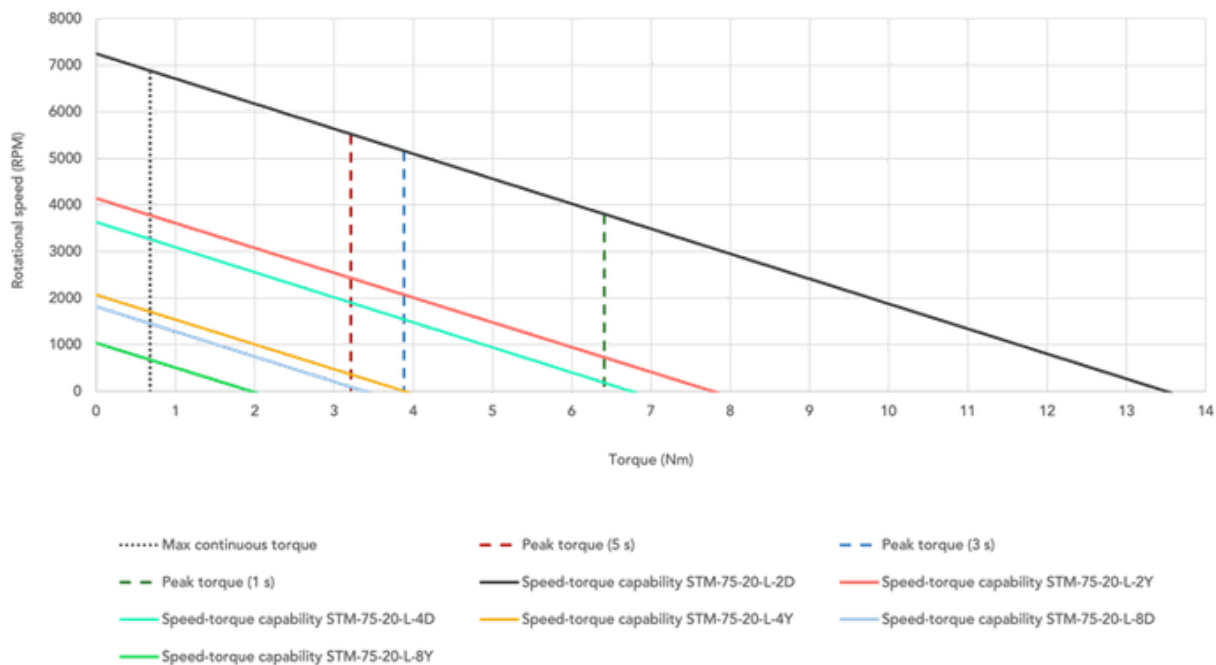
Lite



Max

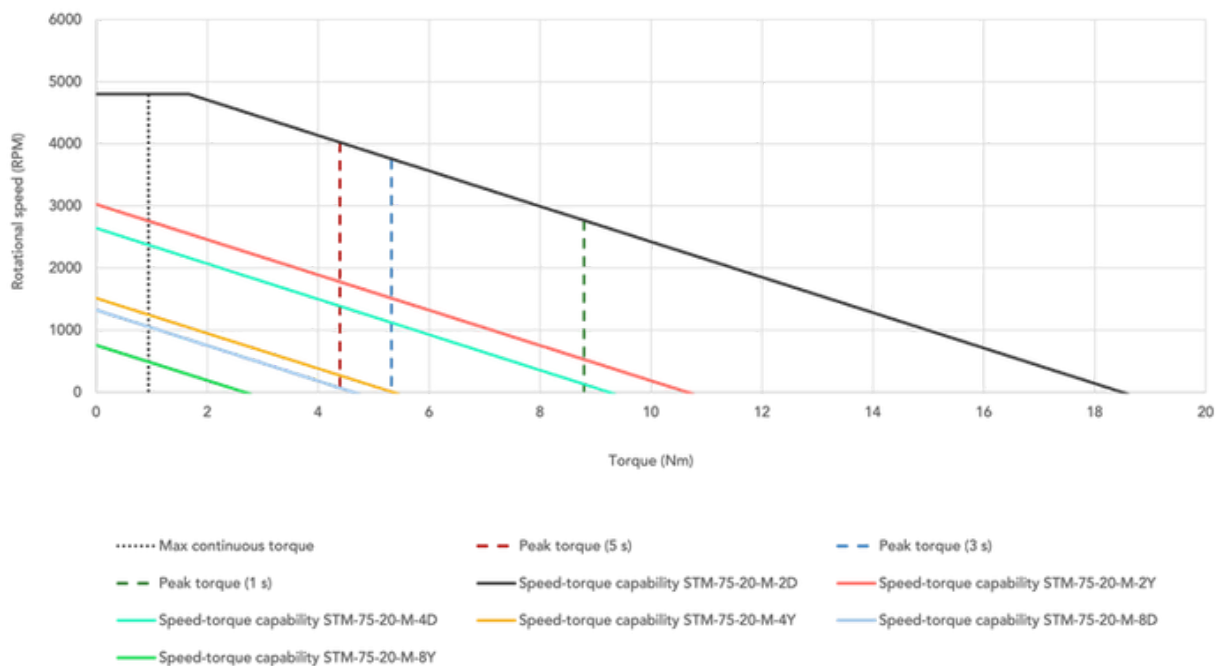


Speed-Torque - STM-75-20 Lite @20 V



*Speed-Torque curves can be generated for your specific supply voltage, please contact sales@alvaindustries.com if you require this.

Speed-Torque curves - STM-75-20 Max @20 V

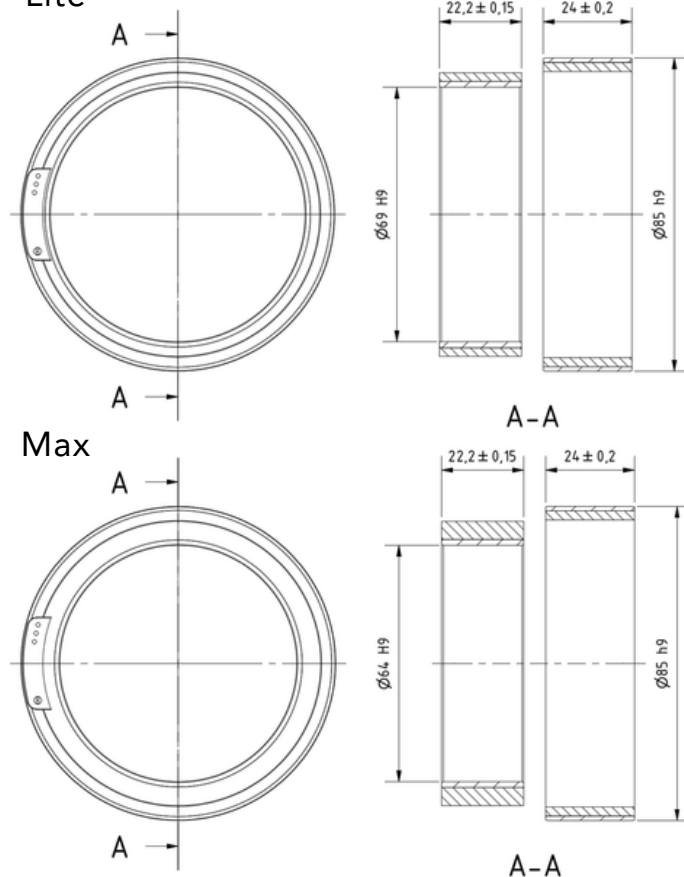


SlimTorq™ 85-24

PROPERTY	UNIT	STM-85-24-L						STM-85-24-M					
Motor type	-	Lite						Max					
Winding type	-	2D	2Y	4D	4Y	8D	8Y	2D	2Y	4D	4Y	8D	8Y
PERFORMANCE													
Motor constant ¹	Nm/√W	0.188						0.252					
Continuous torque ²	Nm	1.06						1.44					
Peak torque, 1 s ³	Nm	7.98						10.7					
Peak torque, 3 s ³	Nm	5.64						7.57					
Peak torque, 5 s ³	Nm	4.84						6.49					
Cogging torque	Nm	0.0						0.0					
Max continuous current ²	Arms	17.5	10.1	8.7	5.0	4.4	2.5	17.7	10.2	8.9	5.1	4.4	2.6
Peak current, 1 s ³	Arms	131	75.8	65.6	37.9	32.8	18.9	131	75.8	65.6	37.9	32.8	18.9
Peak current, 3 s ³	Arms	92.7	53.5	46.4	26.8	23.2	13.4	92.7	53.5	46.4	26.8	23.2	13.4
Peak current, 5 s ³	Arms	79.6	45.9	39.8	23	19.9	11.5	79.6	45.9	39.8	23	19.9	11.5
Max speed at max voltage ⁴	RPM	7900	7900	6960	3970	3480	1990	5100	5100	5100	2970	2590	1480
ELECTRICAL													
Torque constant ¹	Nm/Arms	0.061	0.105	0.122	0.211	0.243	0.421	0.082	0.141	0.163	0.283	0.326	0.566
Voltage constant ¹	V/kRPM	5.17	9.06	10.3	18.1	20.7	36.2	6.95	12.1	13.9	24.3	27.8	48.6
Line to line resistance ¹	Ohm	0.070	0.21	0.28	0.84	1.12	3.36	0.070	0.21	0.28	0.84	1.12	3.36
Line to line inductance ¹	μH	1.9	5.69	7.59	22.8	30.4	91.1	1.9	5.69	7.59	22.8	30.4	91.1
Electrical time constant	μs	27.1											
Spatial harmonic torque ripple	%	0.57						0.43					

PROPERTY	UNIT	STM-85-24-L	STM-85-24-M
Rated voltage	V	72	
Poles	-	42	
Thermal resistance ²	C/W	1.98	
Max stator temperature	°C	140	
Max rotor temperature	°C	85	
Continuous power loss ²	W	54.9	55.6
MECHANICAL			
Stator outer diameter (OD)	mm	85	85
Stator OD tolerance	mm	h9	h9
Rotor inner diameter (ID)	mm	69	64
Rotor ID tolerance	mm	H9	H9
Stator axial length	mm	24	24
Stator axial length tolerance	mm	+/- 0.2	+/- 0.2
Rotor axial length	mm	22.2	22.2
Rotor axial length tolerance	mm	+/- 0.15	+/- 0.15
Rotational inertia	gcm ²	1380	2480
Stator mass ⁵	g	98.4	98.4
Rotor mass	g	101	192
Total mass ⁵	g	200	291
Power cable length	mm	500	500
Power cable size	-	AWG 24	AWG 24
ENVIRONMENTAL			
Minimum ambient temperature	°C	-40	
Maximum ambient temperature	°C	80	
Ambient conditions	°C	20	

Lite

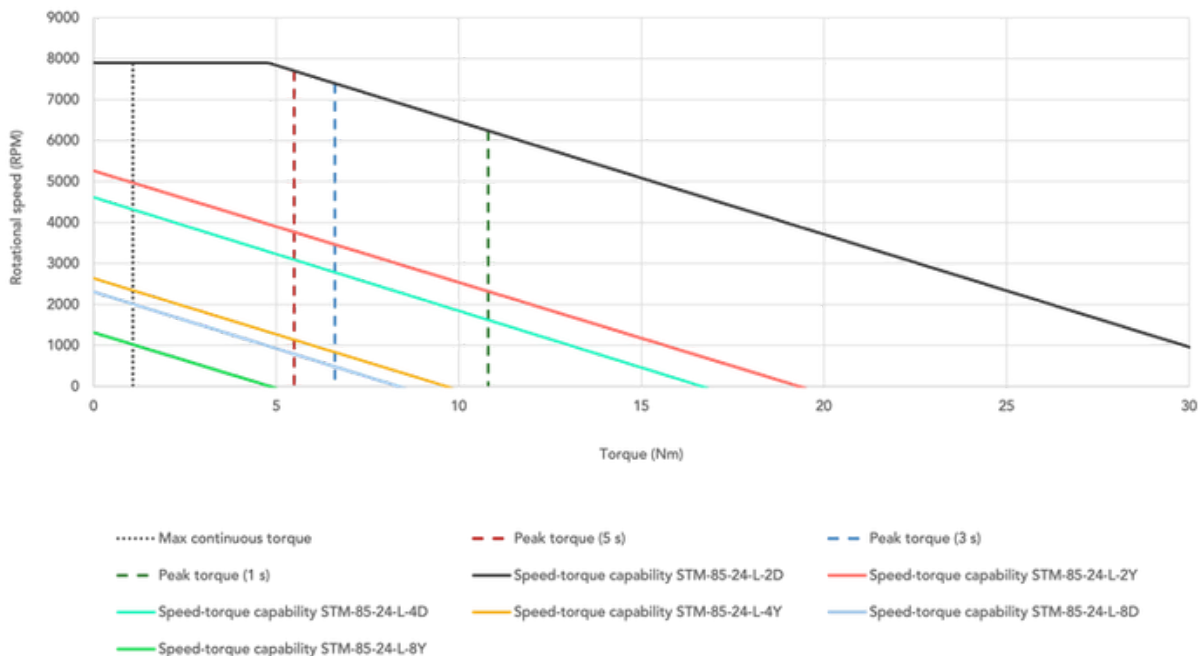


¹Windings at 20 °C & w/o AC cables. ²Housing without fins, 5 m/s airflow, 20°C ambient temp (conservative cooling assumptions, contact us for a continuous torque analysis for your application).

³Winding temperature increases from 20 °C to 140 °C. ⁴Lowest of max mechanical speed & max electrical speed at rated voltage. ⁵W/o AC cables.

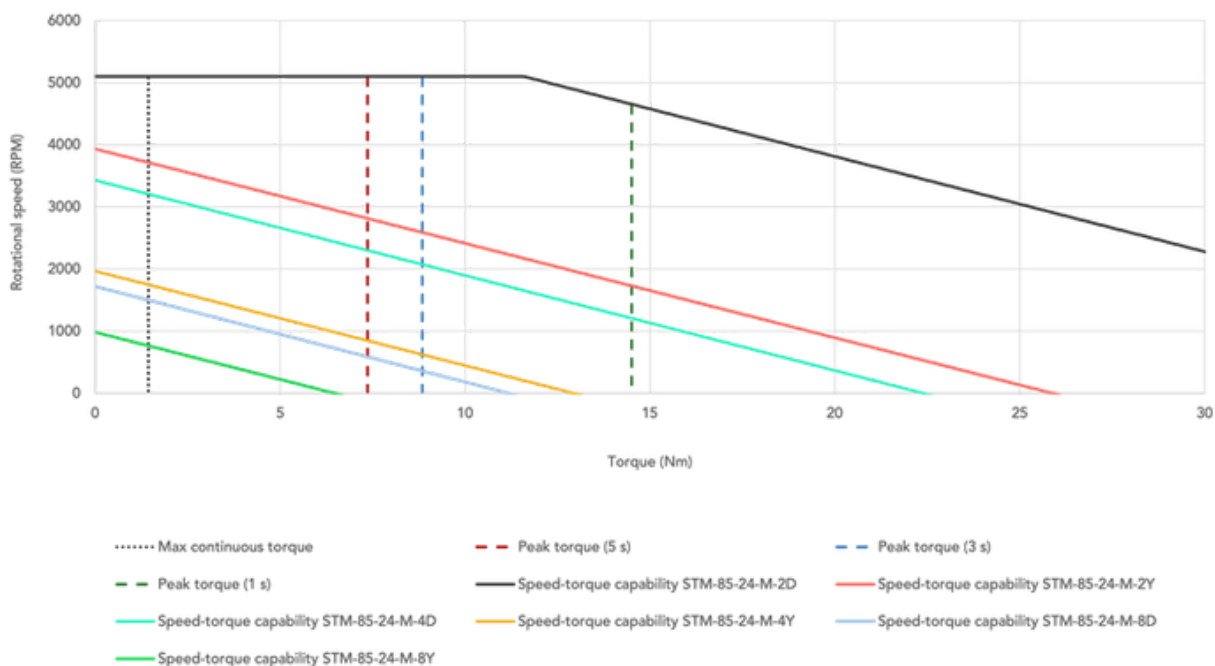
Disclaimer: The values & specifications provided are for reference only & may be subject to change or contain inaccuracies.

Speed-Torque curves - STM-85-24 Lite @48 V



*Speed-Torque curves can be generated for your specific supply voltage, please contact sales@alvaindustries.com if you require this.

Speed-Torque curves - STM-85-24 Max @48 V

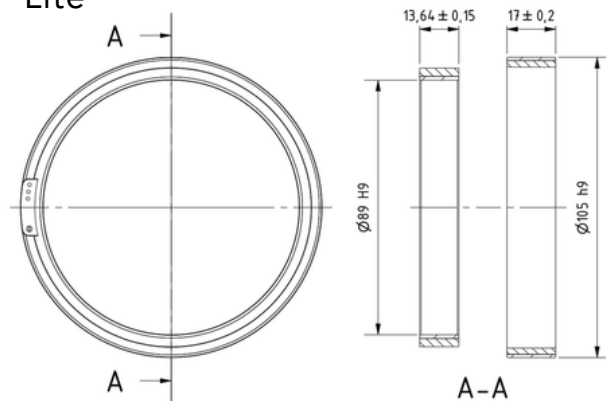


SlimTorq™ 105-17

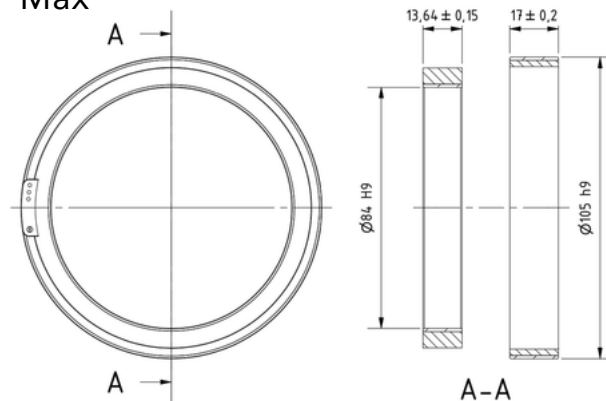
PROPERTY	UNIT	STM-105-17-L						STM-105-17-M					
Motor type	-	Lite						Max					
Winding type	-	2D	2Y	4D	4Y	8D	8Y	2D	2Y	4D	4Y	8D	8Y
PERFORMANCE													
Motor constant ¹	Nm/√W	0.214						0.259					
Continuous torque ²	Nm	1.14						1.39					
Peak torque, 1 s ³	Nm	8.61						10.5					
Peak torque, 3 s ³	Nm	6.22						7.56					
Peak torque, 5 s ³	Nm	5.33						6.47					
Cogging torque	Nm	0.0						0.0					
Max continuous current ²	Arms	15.3	8.8	7.7	4.4	3.8	2.2	15.4	8.9	7.7	4.4	3.9	2.2
Peak current, 1 s ³	Arms	116	67	58.1	33.5	29	16.8	116	67	58.1	33.5	29	16.8
Peak current, 3 s ³	Arms	83.9	48.5	42	24.2	21	12.1	83.9	48.5	42	24.2	21	12.1
Peak current, 5 s ³	Arms	71.9	41.5	35.9	20.8	18.0	10.4	71.9	41.5	35.9	20.8	18.0	10.4
Max speed at max voltage ⁴	RPM	6200	6200	5710	3260	2850	1630	4300	4300	4300	2690	2350	1340
ELECTRICAL													
Torque constant ¹	Nm/Arms	0.074	0.128	0.148	0.257	0.297	0.514	0.090	0.156	0.180	0.312	0.36	0.624
Voltage constant ¹	V/kRPM	6.31	11	12.6	22.1	25.2	44.1	7.67	13.4	15.3	26.8	30.7	53.5
Line to line resistance ¹	Ohm	0.080	0.241	0.321	0.964	1.29	3.86	0.080	0.241	0.321	0.964	1.29	3.86
Line to line inductance ¹	μH	1.73	5.20	6.93	20.8	27.7	83.2	1.73	5.20	6.93	20.8	27.7	83.2
Electrical time constant	μs	21.6											
Spatial harmonic torque ripple	%	0.50						0.40					

PROPERTY	UNIT	STM-105-17-L	STM-105-17-M
Rated voltage	V	72	
Poles	-	54	
Thermal resistance ²	C/W	2.24	
Max stator temperature	°C	140	
Max rotor temperature	°C	85	
Continuous power loss ²	W	48.6	48.9
MECHANICAL			
Stator outer diameter (OD)	mm	105	105
Stator OD tolerance	mm	h9	h9
Rotor inner diameter (ID)	mm	89	84
Rotor ID tolerance	mm	H9	H9
Stator axial length	mm	17	17
Stator axial length tolerance	mm	+/- 0.2	+/- 0.2
Rotor axial length	mm	13.6	13.6
Rotor axial length tolerance	mm	+/- 0.15	+/- 0.15
Rotational inertia	gcm ²	2070	3550
Stator mass ⁵	g	82.4	82.4
Rotor mass	g	93.9	168
Total mass ⁵	g	177	250
Power cable length	mm	500	500
Power cable size	-	AWG 22	AWG 22
ENVIRONMENTAL			
Minimum ambient temperature	°C	-40	
Maximum ambient temperature	°C	80	
Ambient conditions	°C	20	

Lite



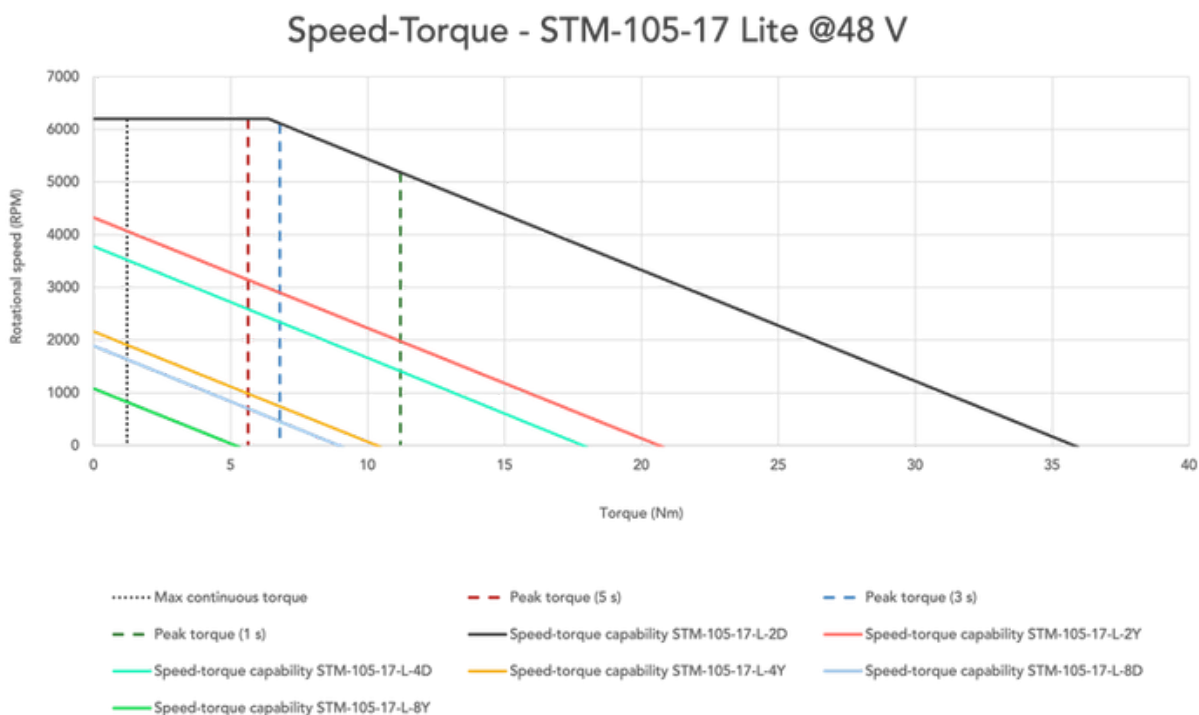
Max



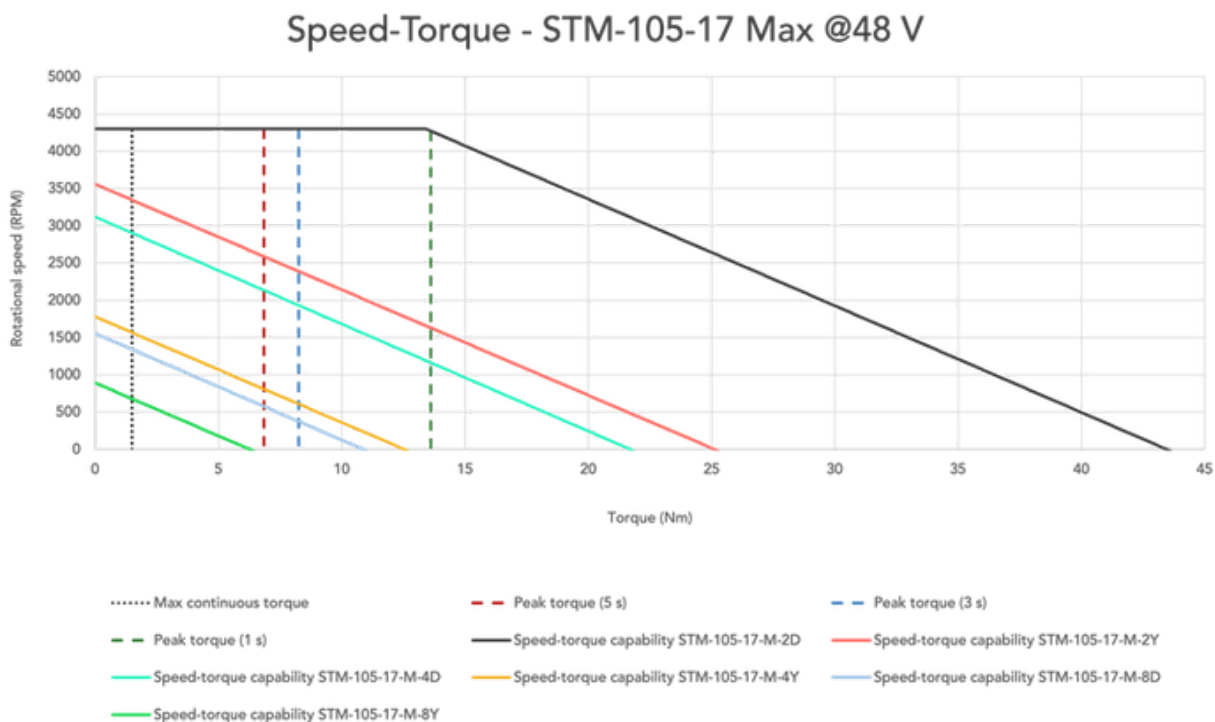
¹Windings at 20 °C & w/o AC cables. ²Housing without fins, 5 m/s airflow, 20°C ambient temp (conservative cooling assumptions, contact us for a continuous torque analysis for your application).

³Winding temperature increases from 20 °C to 140 °C. ⁴Lowest of max mechanical speed & max electrical speed at rated voltage. ⁵W/o AC cables.

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SlimTorq™ 130-27

PROPERTY	UNIT	STM-130-27-L						STM-130-27-M					
Motor type	-	Lite						Max					
Winding type	-	2D	2Y	4D	4Y	8D	8Y	2D	2Y	4D	4Y	8D	8Y
PERFORMANCE													
Motor constant ¹	Nm/√W	0.296						0.507					
Continuous torque ²	Nm	2.03						3.51					
Peak torque, 1 s ³	Nm	19.7						33.8					
Peak torque, 3 s ³	Nm	13.00						22.4					
Peak torque, 5 s ³	Nm	11.00						18.80					
Cogging torque	Nm	0.0						0.0					
Max continuous current ²	Arms	31.1	18.0	15.6	9.0	7.8	4.5	31.4	18.1	15.7	9.1	7.9	4.5
Peak current, 1 s ³	Arms	303	175	151	87.4	75.7	43.7	303	175	151	87.4	75.7	43.7
Peak current, 3 s ³	Arms	200	116	100	57.8	50.1	28.9	200	116	100	57.8	50.1	28.9
Peak current, 5 s ³	Arms	169	97.3	84.3	48.7	42.1	24.3	169	97.3	84.3	48.6	42.1	24.3
Max speed at max voltage ⁴	RPM	6300	6300	6300	3700	3260	1850	4200	4200	3800	2160	1900	1080
ELECTRICAL													
Torque constant ¹	Nm/Arms	0.065	0.113	0.130	0.226	0.260	0.451	0.112	0.193	0.223	0.387	0.446	0.773
Voltage constant ¹	V/kRPM	5.52	9.73	11	19.5	22.1	38.9	9.48	16.6	19	33.3	37.9	66.5
Line to line resistance ¹	Ohm	0.032	0.0968	0.129	0.387	0.516	1.55	0.032	0.0968	0.129	0.387	0.516	1.55
Line to line inductance ¹	μH	2	6.00	8	24	32	96	2	6.00	8	24	32	96
Electrical time constant	μs	62											
Spatial harmonic torque ripple	%	0.92						0.62					

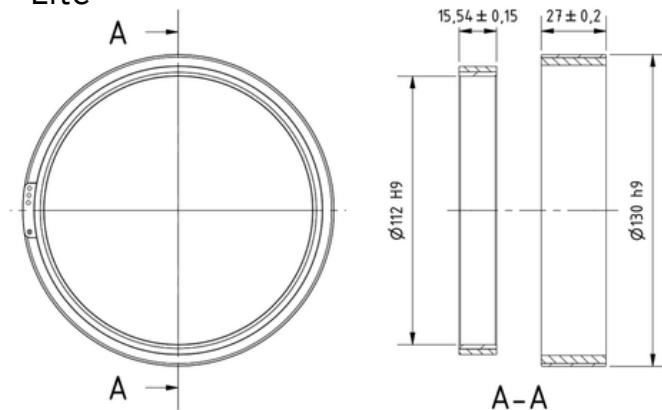
PROPERTY	UNIT	STM-130-27-L	STM-130-27-M
Rated voltage	V	72	
Poles	-	46	
Thermal resistance ²	C/W	1.35	
Max stator temperature	°C	140	
Max rotor temperature	°C	85	
Continuous power loss ²	W	80.7	81.4
MECHANICAL			
Stator outer diameter (OD)	mm	130	130
Stator OD tolerance	mm	h9	h9
Rotor inner diameter (ID)	mm	112	107
Rotor ID tolerance	mm	H9	H9
Stator axial length	mm	27	27
Stator axial length tolerance	mm	+/- 0.2	+/- 0.2
Rotor axial length	mm	15.5	18.3
Rotor axial length tolerance	mm	+/- 0.15	+/- 0.15
Rotational inertia	gcm ²	3910	8360
Stator mass ⁵	g	221	221
Rotor mass	g	116	254
Total mass ⁵	g	337	475
Power cable length	mm	500	500
Power cable size	-	AWG 18	AWG 18
ENVIRONMENTAL			
Minimum ambient temperature	°C	-40	
Maximum ambient temperature	°C	80	
Ambient conditions	°C	20	

¹Windings at 20 °C & w/o AC cables. ²Housing without fins, 5 m/s airflow, 20°C ambient temp (conservative cooling assumptions, contact us for a continuous torque analysis for your application).

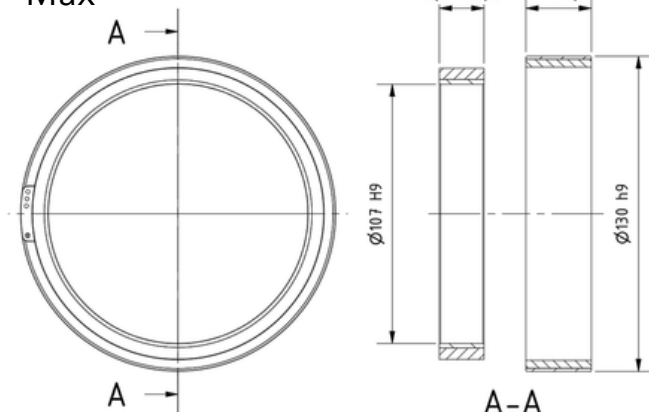
³Winding temperature increases from 20 °C to 140 °C. ⁴Lowest of max mechanical speed & max electrical speed at rated voltage. ⁵W/o AC cables.

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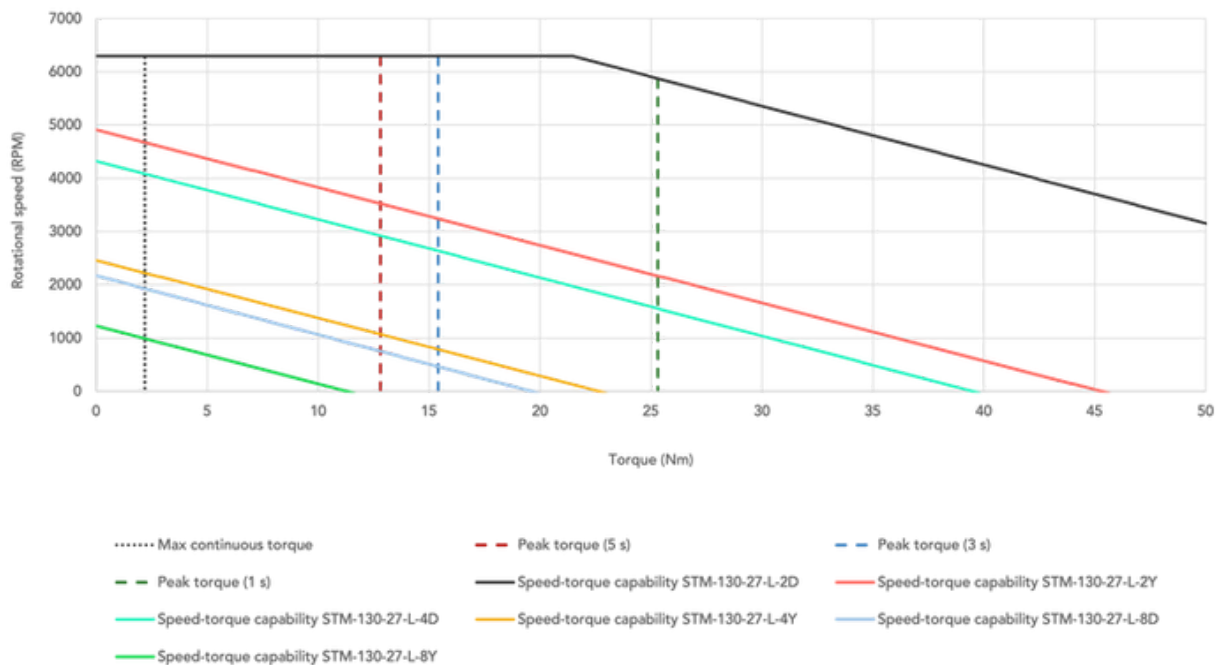
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Max

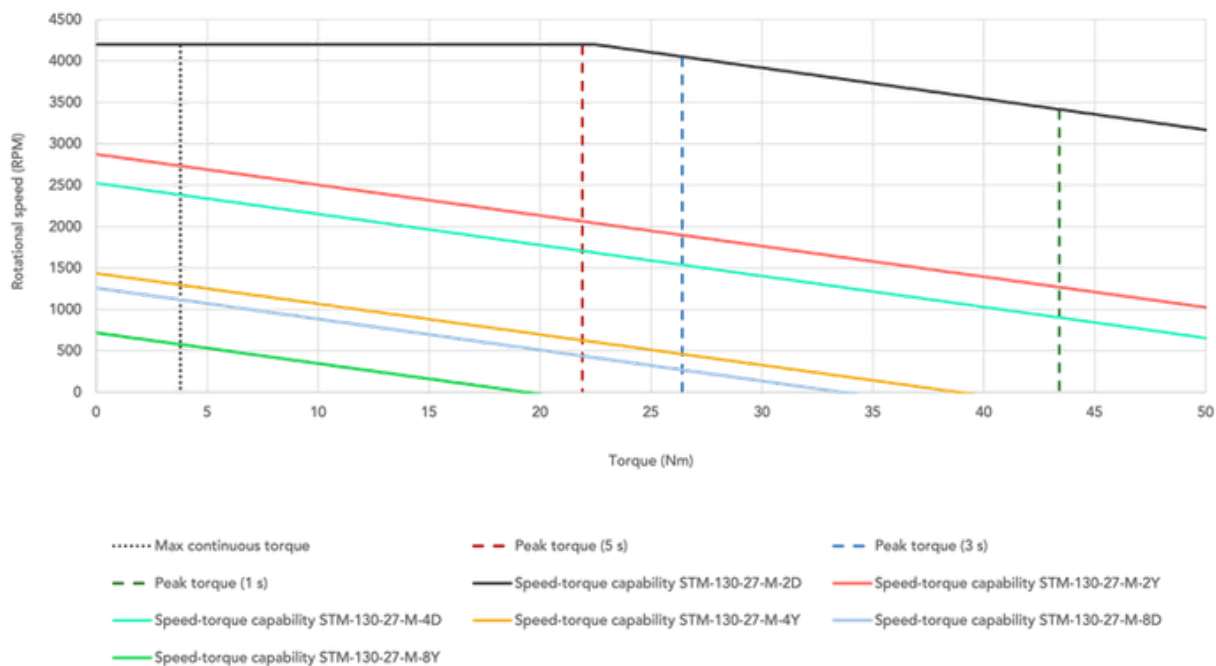


Speed-Torque - STM-130-27 Lite @48 V



*Speed-Torque curves can be generated for your specific supply voltage,
please contact sales@alvaindustries.com if you require this.

Speed-Torque - STM-130-27 Max @48 V

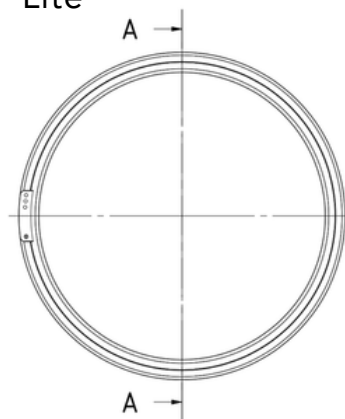


SlimTorq™ 160-17

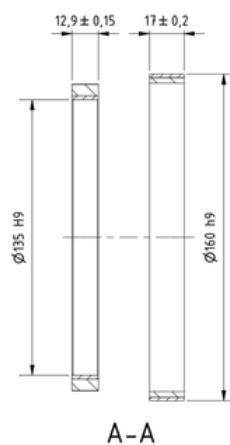
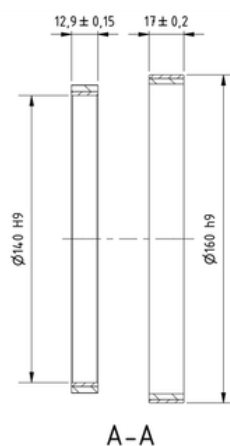
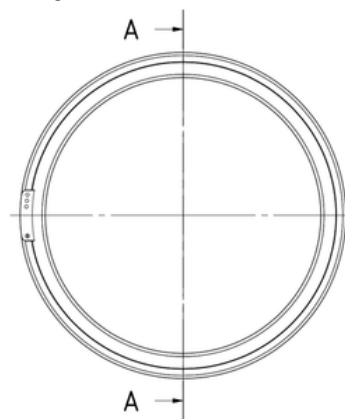
PROPERTY	UNIT	STM-160-17-L						STM-160-17-M					
Motor type	-	Lite						Max					
Winding type	-	2D	2Y	4D	4Y	8D	8Y	2D	2Y	4D	4Y	8D	8Y
PERFORMANCE													
Motor constant ¹	Nm/√W	0.377						0.469					
Continuous torque ²	Nm	2.34						2.94					
Peak torque, 1 s ³	Nm	18.8						23.4					
Peak torque, 3 s ³	Nm	13.60						16.9					
Peak torque, 5 s ³	Nm	11.80						14.60					
Cogging torque	Nm	0.0						0.0					
Max continuous current ²	Arms	17.3	10.0	8.7	5.0	4.3	2.5	17.5	10.1	8.8	5.1	4.4	2.5
Peak current, 1 s ³	Arms	139	80.4	69.6	40.2	34.8	20.1	139	80.4	69.6	40.2	34.8	20.1
Peak current, 3 s ³	Arms	101	58.1	50.3	29	25.1	14.5	101	58.1	50.3	29	25.1	14.5
Peak current, 5 s ³	Arms	87.1	50.3	43.6	25.1	21.8	12.6	87.1	50.3	43.6	25.1	21.8	12.6
Max speed at max voltage ⁴	RPM	4800	3580	3140	1790	1570	895	3400	2880	2520	1440	1260	721
ELECTRICAL													
Torque constant ¹	Nm/Arms	0.135	0.234	0.270	0.467	0.540	0.935	0.168	0.291	0.336	0.581	0.671	1.16
Voltage constant ¹	V/kRPM	11.5	20.1	22.9	40.2	45.9	80.4	14.3	25	28.5	50	57.1	99.9
Line to line resistance ¹	Ohm	0.085	0.256	0.341	1.02	1.36	4.09	0.085	0.256	0.341	1.02	1.36	4.09
Line to line inductance ¹	μH	2.19	6.57	8.76	26.3	35	105	2.19	6.57	8.76	26.3	35	105
Electrical time constant	μs	25.7											
Spatial harmonic torque ripple	%	0.64						0.51					

PROPERTY	UNIT	STM-160-17-L	STM-160-17-M
Rated voltage	V	72	
Poles	-	68	
Thermal resistance ²	C/W	1.65	
Max stator temperature	°C	140	
Max rotor temperature	°C	85	
Continuous power loss ²	W	73.5	73.6
MECHANICAL			
Stator outer diameter (OD)	mm	160	160
Stator OD tolerance	mm	h9	h9
Rotor inner diameter (ID)	mm	140	135
Rotor ID tolerance	mm	H9	H9
Stator axial length	mm	17	17
Stator axial length tolerance	mm	+/- 0.2	+/- 0.2
Rotor axial length	mm	12.9	12.9
Rotor axial length tolerance	mm	+/- 0.15	+/- 0.15
Rotational inertia	gcm ²	8400	13800
Stator mass ⁵	g	153	153
Rotor mass	g	165	273
Total mass ⁵	g	318	426
Power cable length	mm	500	500
Power cable size	-	AWG 18	AWG 18
ENVIRONMENTAL			
Minimum ambient temperature	°C	-40	
Maximum ambient temperature	°C	80	
Ambient conditions	°C	20	

Lite



Max

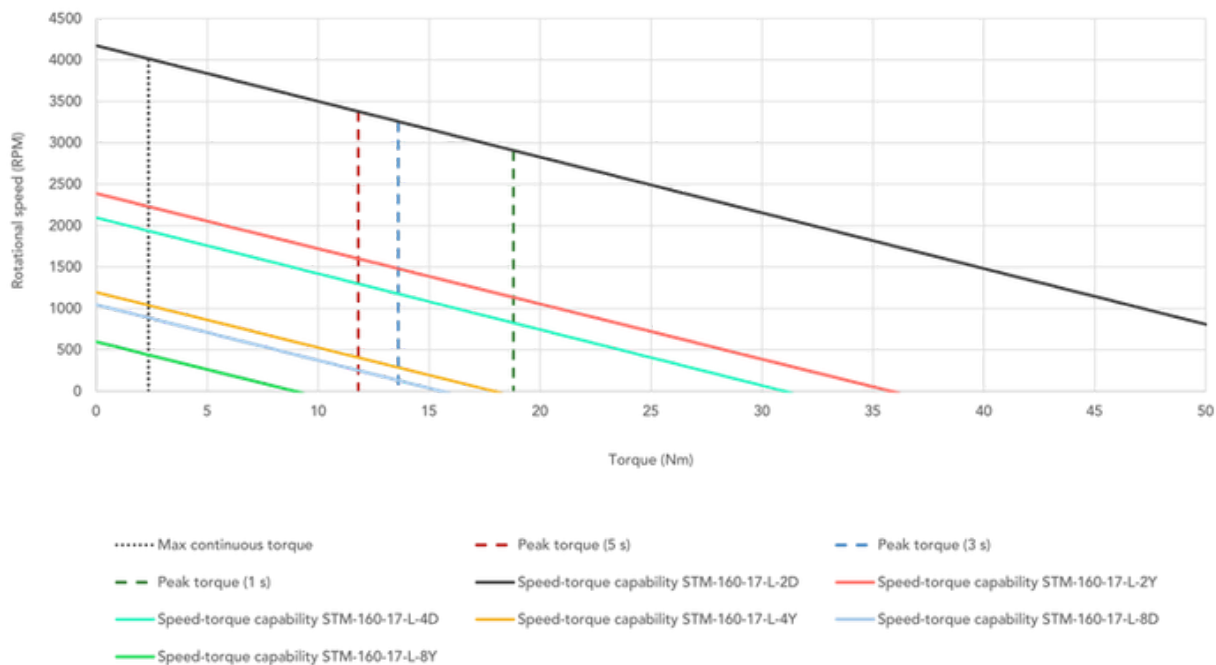


¹Windings at 20 °C & w/o AC cables. ²Housing without fins, 5 m/s airflow, 20°C ambient temp (conservative cooling assumptions, contact us for a continuous torque analysis for your application).

³Winding temperature increases from 20 °C to 140 °C. ⁴Lowest of max mechanical speed & max electrical speed at rated voltage. ⁵W/o AC cables.

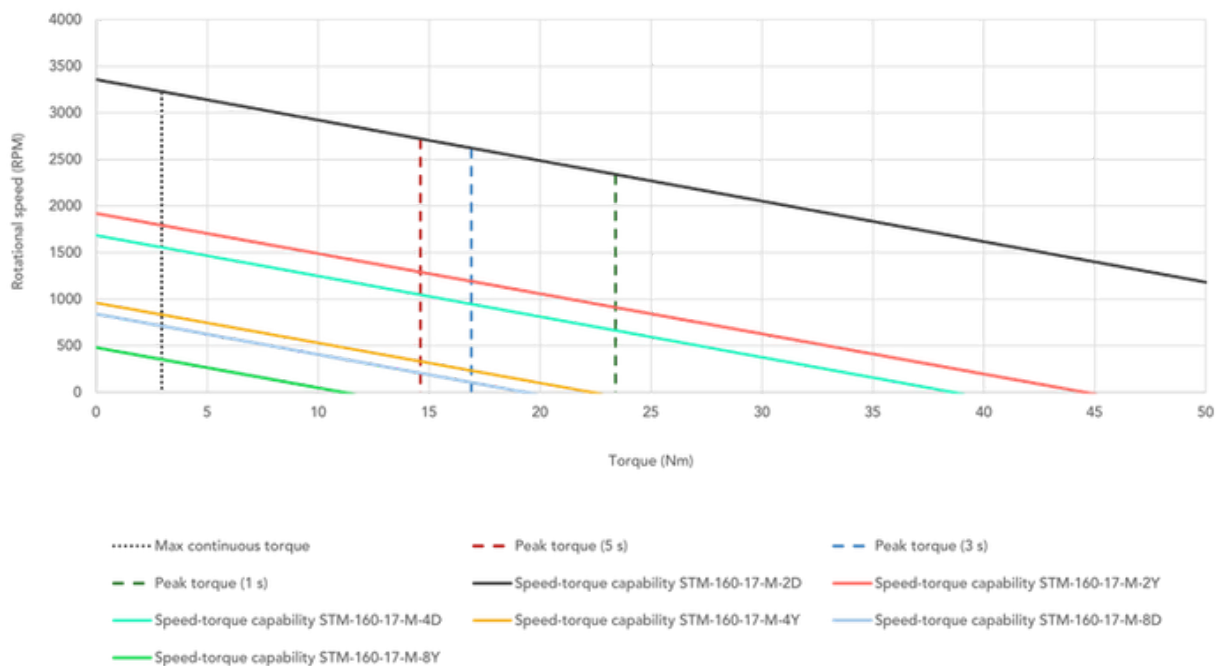
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Speed-Torque - STM-160-17 Lite @48 V



*Speed-Torque curves can be generated for your specific supply voltage,
please contact sales@alvaindustries.com if you require this.

Speed-Torque - STM-160-17 Max @48 V

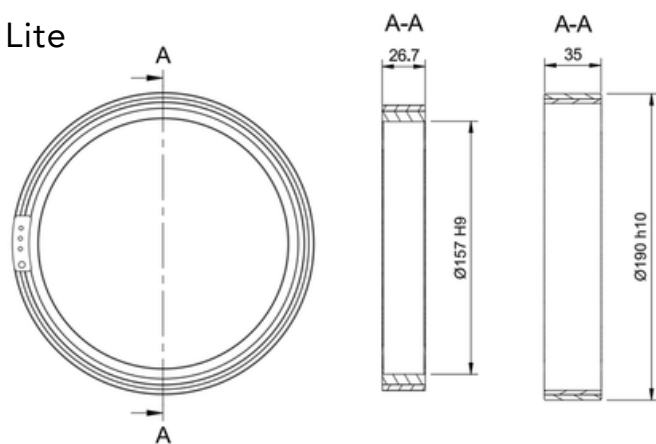


SlimTorq™ 190-35

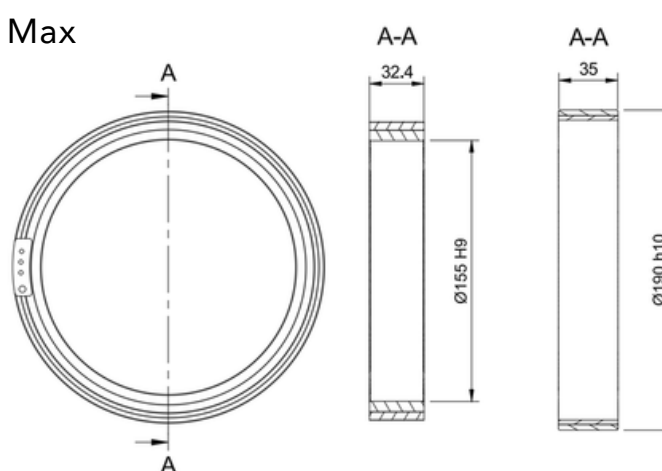
PROPERTY	UNIT	STM-190-35-L						STM-190-35-M					
Motor type	-	Lite						Max					
Winding type	-	2D	2Y	4D	4Y	8D	8Y	2D	2Y	4D	4Y	8D	8Y
PERFORMANCE													
Motor constant ¹	Nm/√W	0.84						1.08					
Continuous torque ²	Nm	8.72						11.29					
Peak torque, 1 s ³	Nm	68.31						88.05					
Peak torque, 3 s ³	Nm	48.17						62.10					
Peak torque, 5 s ³	Nm	41.87						53.98					
Cogging torque	Nm	0.0						0.0					
Max continuous current ²	Arms	28.6	16.5	14.3	8.3	7.2	4.1	28.6	16.5	14.3	8.3	7.2	4.1
Peak current, 1 s ³	Arms	207.6	119.8	103.8	59.9	51.9	30.0	200.6	115.9	100.3	57.9	50.2	29.0
Peak current, 3 s ³	Arms	146.4	84.5	73.2	42.3	36.6	21.1	141.5	81.7	70.7	40.9	35.4	20.4
Peak current, 5 s ³	Arms	127.2	73.4	63.6	36.7	31.8	18.4	123.0	71.0	61.5	35.5	30.8	17.8
Max speed at max voltage ⁴	RPM	1900	1464	1286	732	643	366	1300	1136	996	568	498	284
ELECTRICAL													
Torque constant ¹	Nm/Arms	0.329	0.570	0.658	1.140	1.317	2.280	0.439	0.760	0.878	1.520	1.755	3.040
Voltage constant ¹	V/kRPM	28.0	49.0	55.9	98.1	111.9	196.1	37.3	65.3	74.7	130.6	149.3	261.3
Line to line resistance ¹	Ohm	0.105	0.314	0.418	1.254	1.672	5.017	0.105	0.314	0.418	1.254	1.672	5.017
Line to line inductance ¹	μH	5	15	20	59	79	237	5	15	20	59	79	237
Electrical time constant	μs	47.2											
Spatial harmonic torque ripple	%	0.59						0.51					

PROPERTY	UNIT	STM-190-35-L	STM-190-35-M
Rated voltage	V	72	
Poles	-	74	
Thermal resistance ¹	C/W	0.57	
Max stator temperature	°C	140	
Max rotor temperature	°C	85	
Continuous power loss ¹	W	189.2	189.2
MECHANICAL			
Stator outer diameter (OD)	mm	190	190
Stator OD tolerance	mm	h10	h10
Rotor inner diameter (ID)	mm	157	155
Rotor ID tolerance	mm	H9	H9
Stator axial length	mm	35	35
Stator axial length tolerance	mm	+0.2	+0.2
Rotor axial length	mm	26.7	32.4
Rotor axial length tolerance	mm	+0.15	+0.15
Rotational inertia	gcm ²	40042	59538
Stator mass ⁵	g	489	489
Rotor mass	g	551	810
Total mass ⁵	g	1089	1352
Power cable length	mm	500	500
Power cable size	-	AWG16	AWG16
ENVIRONMENTAL			
Minimum ambient temperature	°C	-40	
Maximum ambient temperature	°C	80	
Ambient conditions	°C	20	

Lite



Max

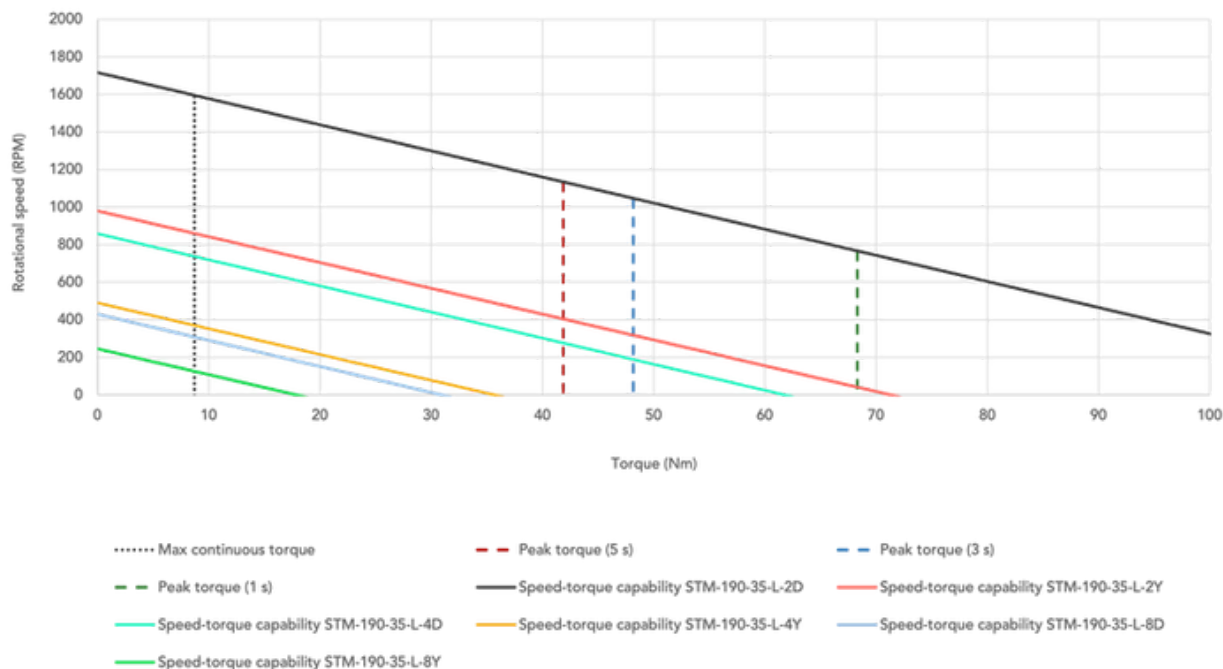


¹Windings at 20 °C & w/o AC cables. ²Housing without fins, 5 m/s airflow, 20°C ambient temp (conservative cooling assumptions, contact us for a continuous torque analysis for your application).

³Winding temperature increases from 20 °C to 140 °C. ⁴Lowest of max mechanical speed & max electrical speed at rated voltage. ⁵W/o AC cables.

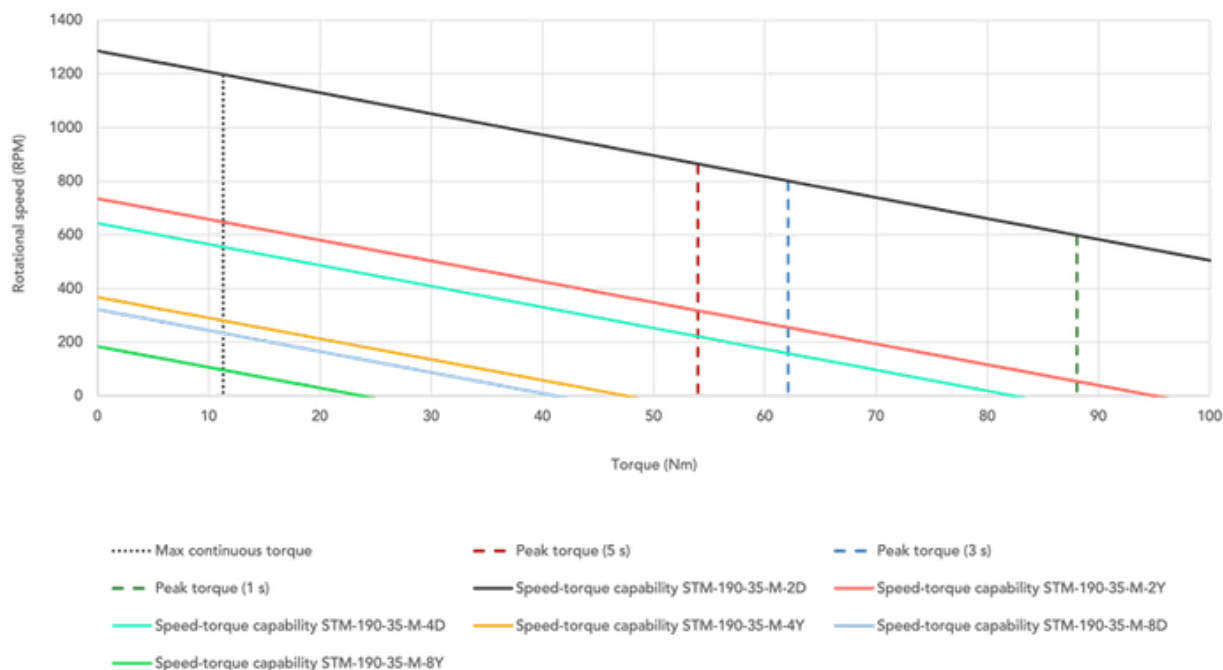
Disclaimer: The values & specifications provided are for reference only & may be subject to change or contain inaccuracies.

Speed-Torque - STM-190-35 Lite @48 V



*Speed-Torque curves can be generated for your specific supply voltage, please contact sales@alvaindustries.com if you require this.

Speed-Torque - STM-190-35 Max @48 V



Ordering Alva Motors

Pricing & Lead-Time

Alva's pricing strategy focuses on delivering customer value while ensuring sustainable profitability. Instead of cost-plus pricing, we collaborate with customers to establish pricing that fosters win-win outcomes. Our goal is to maintain stable pricing, allowing customers to rely on consistent quality. Final pricing will depend on factors like part configuration, motor size, annual volumes, & testing requirements.

Standard products are delivered in 8-10 weeks for small quantities, with plans to reduce this to 4-6 weeks. Custom solutions may extend lead times to 12-16 weeks, while full customization can take up to 20 weeks from specifications to prototype. We work closely with customers to meet timelines, & volume lead times are determined upon request based on customer needs.

Delivery Terms

Alva's standard delivery terms are EXW; however, Alva remains flexible & open to discussing customized delivery arrangements to meet customers' specific needs.

Terms & Conditions

Alva's term & conditions can be found under www.alvaindustries.com/terms-conditions & as an annex of this product guide.

How to place an order

Use the [Product Selector](#) to configure & order your motor or get in touch with Alva's sales team at sales@alvaindustries.com to receive an official quotation according to which an order can be placed, or a sales contract can be signed.

Scope of delivery

Stator
Fiberprinted™ stator with a aluminium sleeve with tight tolerances for easy mechanical integration. The aluminium sleeve is engraved with the PN & SN of the stator.

The stator comes with pre-installed cables with a length of 500mm with flying leads. The standard cable is UL rated with PTFE insulation material.

Rotor
The rotor consists of a magnetic Halbach, for maximum performance with a glass fiber sleeve, for maximum reliability, and a inner aluminium ring with tight tolerances for easy mechanical integration.

The inner aluminium ring is engraved with the PN & SN of the rotor.

STM-130-27-L-4Y-A-18A-A0-001

Motor Series
STM - SlimTorq™ Motor

Diameter
Stator OD (mm)

Axial Length
Stator Length (mm)

Motor Variant
L - Lite
M - Max

Unique Identifier

Sensor Options
Temperature + Position (A-Z) + (A-Z) (0 = None)

Cable Gauge & Type
AWG(1-99) + Type(A-Z)

Terminal Option
A - Axial Direction
R - Radial Direction

Winding Option
Series Turns + Star(Y)/Delta(D)

Evaluation Kits



High-end Bearings

Aluminium Housing

RLS AksIM-2™



Kick-start your testing

SlimTorq™ motors can be ordered as evaluation kits, which includes a SlimTorq™ frameless motor, bearings, & a RLS AksIM-2™ (BiSS-C) encoder; all integrated into an aluminium housing for easy setup and use.

Easy mounting

The framed motors comes with a mechanical interface adapter to simplify the installation into your test-bench.

CAD-files of the framed motors are available on our [website](#).

Motor	Encoder
STM-39-10	RLS: MRA029GP013DMN00
	RLS: MB029DCC18BFNT00
STM-51-12	RLS: MRA049BC025DSE00
	RLS: MB049DCC17BDNT00
STM-75-20	RLS: MRA064BC040DSE00
	RLS: MB064DCC20BDNT00
STM-85-24	RLS: MRA064BC040DSE00
	RLS: MB064DCC20BDNT00
STM-105-17	RLS: MRA064BC040DSE00
	RLS: MB064DCC20BDNT00
STM-130-27	RLS: MRA080DF068DMH00
	RLS: MB080DCC20BDNT00



Mechanical Integration

Safety precautions

The frameless motor rotors contain high-energy rare-earth magnets that produce strong magnetic fields. Therefore, anyone with active implants, such as pacemakers, should keep a safe distance from the rotors of these motors.

The rotor's magnetic field creates strong attractive forces to ferromagnetic materials when in close proximity. If the rotor & surrounding ferromagnetic items are not properly restrained, this can result in collisions, potentially damaging the rotor or causing personnel injuries (e.g., pinched fingers). When handling the rotor, precautions should be taken to remove all other magnetic items from an area greater than approximately 3 times the diameter of the rotor. This includes other rotors.

Strong magnetic fields produced by the rotor magnets may affect or damage electronic devices & measuring equipment. Rotors should not be placed near computer disks, computers, monitors, credit cards, etc.



Handling & Storage

Correct handling & storage of motor components is very important. It is recommended that stators & rotors be stored in their original packaging until installation is required. Special care must be taken when handling stator assemblies, as damage to the coil insulation and lead wires can result in electrical shorts and possible electrical shocks.

Precautions should also be taken when handling the rotor assembly, as strong magnetic forces are present. The magnets on the rotor are brittle & prone to cracking, chipping, or even shattering on impact.

Do not pull on the phase cables, while stripping the cables, a proper cable stripping tool that does not pull on the cable must be used.

Mechanical Integration

General considerations

Alva frameless motors are supplied as rotor & stator pairs. From this point forward, the part that the rotor attaches to will be referred to as the rotor carrier, & the part that the stator attaches to will be referred to as the housing. Both the stator & rotor have an aluminium backing ring(sleeve) on the mounting side. Only the exposed surfaces of these rings should be subjected to force during assembly or disassembly. Depending on the motor configuration, it may only be possible to insert the rotor from one side of the stator.



Alva's motors are electromagnetically self-sufficient, i.e., proper operation does not depend on the electromagnetic properties of surrounding parts. For structural purposes, it is recommended that the housing & rotor carrier be made of aluminum. This is particularly beneficial for applications operating within a large temperature range, as the coefficient of thermal expansion (CTE) is matched with the stator/rotor sleeve.

It is good practice to design any electrically conducting surfaces to have a minimum of 0.5mm of clearance to the stator windings in all directions for electrical insulation purposes. The windings should not be in direct contact with any electrically conducting surfaces. If grounding is required, a ground stud can be placed on the housing or stator sleeve. While this may not be at the same voltage potential as the stator stack, it represents a typical grounding scheme used in most motors with two- or three-part housings. A ground stud is typically only necessary when the motor operates without isolation between the mains & the motor drive.

The housing & surrounding parts must be designed with sufficient clearance around the terminal block & phase cables to ensure no stress is applied to the terminal block. Please refer to the STEP files for specific motor measurements & tolerances for the terminal block.

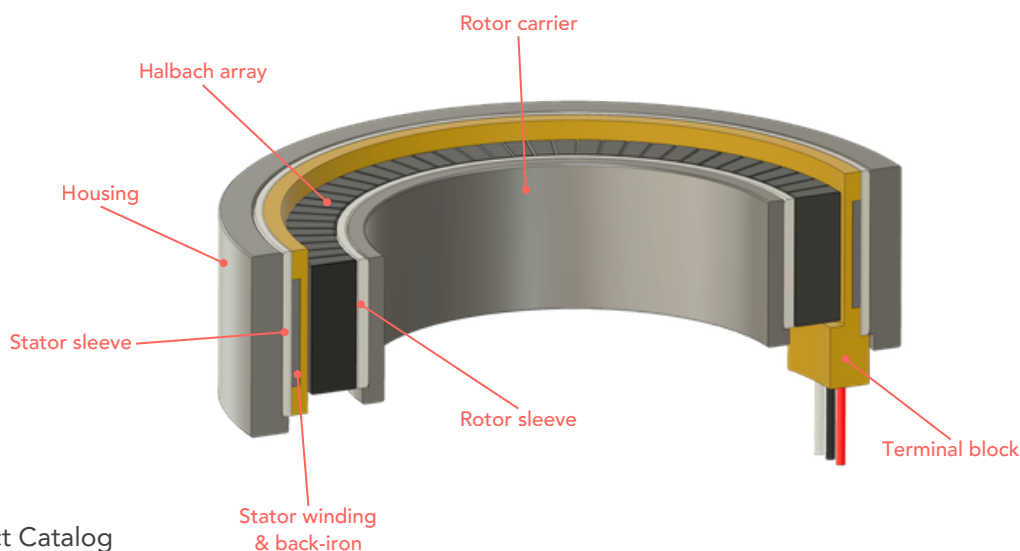
The following section is also available as easy to follow Youtube videos:

[How to Integrate Frameless Motors: SlimTorq™ Motor Assembly Guide](#)

[Axial Clamping Explained: Frameless SlimTorq™ Motor Assembly](#)

[Thin Bond Line Adhesion: Frameless Motor Integration Tutorial](#)

[Thick Bond Line Adhesion: Frameless Motor Integration Tutorial](#)



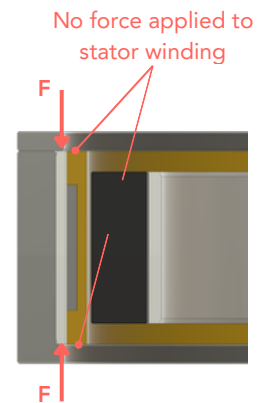
Mechanical Integration

Retention

A well designed integration method is required to assure a strong retention of the rotor & stator to allow for robust torque transfer. The outlined options will all achieve a strong retention if they are applied correctly.

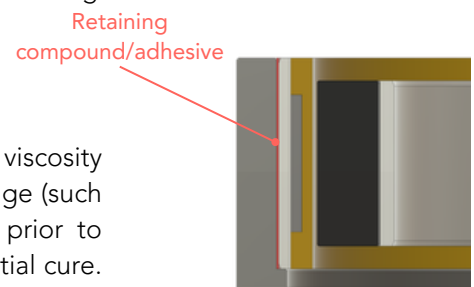
Axial clamping

The simplest method of retention is axial clamping. This allows for easy assembly & disassembly of the stator & rotor. It's recommended to use one or several clamping components to evenly distribute the clamping force on the aluminium sleeves. Clamping force should only be applied to the stator/rotor aluminium sleeve. Absolutely no force should be applied on the stator winding or the rotor magnets. The amount of clamping force as well as the housing shoulder & clamping parts should be designed so that the stresses in the stator/rotor sleeve are appropriate. It's recommended to add a thermal paste on the mounting surfaces to maximise the thermal performance of the motor.



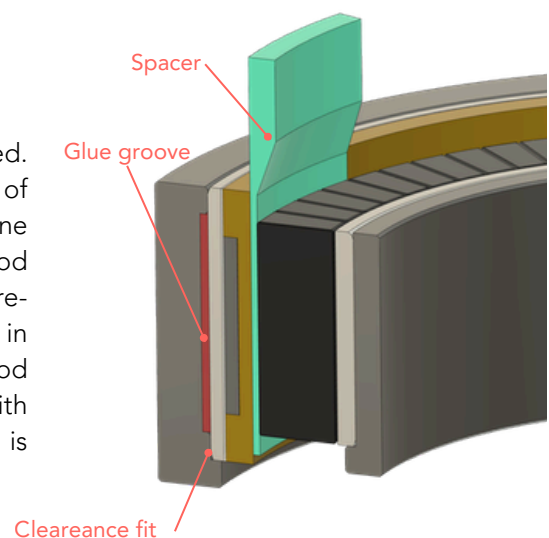
Bonding: Small gap

The most space efficient option is to bond the parts using a low viscosity adhesive or retaining compound with sufficient temperature range (such as Loctite 638). The mating surfaces should be clean & dry prior to assembly & the parts needs to be held in position during the initial cure. It should be noted that disassembly is not possible with this retention method.



Bonding: Large gap

A large gap bonding method can also be used if desired. Stator to rotor alignment can either be achieved by use of spacers (spacers are not included) in the airgap OR combine a clearance fit with a glue groove. Note that spacer method require the rotor (alternatively a dummy rotor) to be pre-mounted on the rotor carrier while the stator is glued in place & vice versa. A general structural adhesive with good gap filling capabilities should be used. In applications with high power demands a thermally conductive adhesive is required (such as the MG chemicals 8329 TFS).



Interference fit

The use of an interference fit to circumvent the need for clamping & adhesives is possible but require careful consideration not to overstress & damage the stator or rotor. Therefore it is not recommended over the use of axial clamping or bonding. Alva should be consulted before a interference fit is implemented.

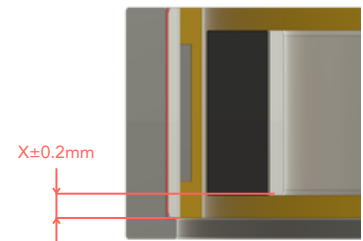
Mechanical Integration

Alignment

A good alignment between the rotor & stator is important to ensure safe operation & high performance of the motor.

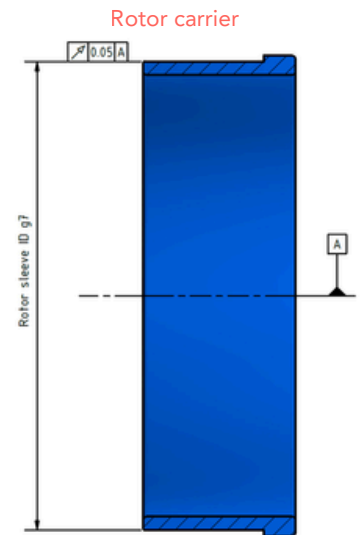
Axial alignment

Axial placement of the stator & rotor should be ensured by the use of a hard-stop shoulder on the rotor carrier & the stator housing. The shoulders should control the axial alignment to center the rotor within the stator with a $\pm 0.2\text{mm}$ tolerance.



Rotor radial alignment

It's recommend to retain the rotor with axial clamping, small bond line, clearance fit with a glue groove or full interference fit. For all the methods except for the full interference fit, it's recommended to dimension the diameter of the rotor carrier (where the rotor ID sits) equal to the rotor sleeve ID with a g7 tolerance, or tighter. It's important that the rotor carrier has maximum runout of 0.05mm relative to the rotation axis if one these methods are used. Please contact Alva if you are considering a full interference fit.

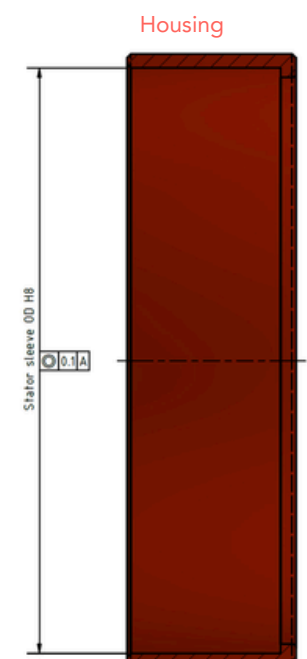


Stator radial alignment

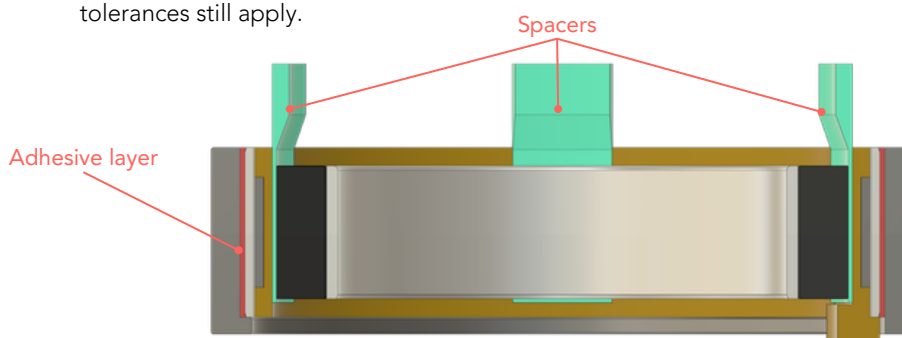
Stator alignment options:

1. Alignment with the stator housing
2. In-situ alignment with spacers

Alignment through the housing is achieved by a clearance fit between the stator sleeve and the housing. In this case the housing mating surface ID should be dimensioned equal to the Stator sleeve OD with a H8 tolerance or tighter. The housing mating surface should be concentric to the rotor carrier rotation axis to within 0.1mm in the final assembly.



In situ alignment is achieved by first assembling the rotor, then bonding the stator into the housing while using spacers between the stator & rotor to ensure the correct airgap. In this case there should be a minimal stator sleeve to housing gap of 0.1mm (minimal, not nominal). This method of assembly, though more process demanding, reduces the need for high precision on the housing ID & general assembly tolerances. Note that the requirement for rotor carrier tolerances still apply.



Choosing a Motor Drive

Commutation Type

SlimTorq™ motors have an almost perfect sinusoidal back-EMF & it is therefore recommended to use sinusoidal, or even better, FOC commutation. Trapezoidal commutation can be utilised, however, with a penalty in terms of precision & performance.

Switching Frequency

Slotless motors have a relatively low inductance, & therefore a low electrical time constant. It's therefore important to use a high enough switching frequency to maximise the precision & performance of the motors in your application. The general recommendation is to use a switching frequency of at least 20kHz. The required switching frequency can be reduced by adding output filter on the motor drive to increase the system inductance. Please contact us at sales@alvaindustries.com if you are uncertain if your application require a output filter.

Current Rating

Most motor drives has two current ratings: continuous- & peak-current. It's important to understand both ratings & assure to pick the motor drive with correct current ratings for your application.

Continuous current

The required continuous current rating can be calculated with the equation below. It's important to consider that the continuous current rating of motor drives is dependant on ambient temperature, cooling conditions & switching frequency. So ensure to review these aspects of the motor drive & if your are uncertain you should contact your motor drive supplier for support.

$$\text{ContinuousCurrent}(A_{rms}) = \frac{\text{ContinuousTorque}(Nm)}{K_T(Nm \div A_{rms})}$$

Electrical Frequency

It's important to select a motor drive that has high enough electrical frequency for your application. It's easy to calculate the required electrical frequency for your application, see the equation below:

$$\text{ElectricalFrequency}(Hz) = \frac{\text{MaxSpeed}(RPM) \times \text{No. Poles}}{2 \times 60(s \div min)}$$

Sensored vs Sensorless control

The SlimTorq™ motors can be controlled both by a sensed & sensorless motor drive. If you have a precision applications where you need to control the motors either in position- or torque-mode it's highly recommended to use a sensed motor drive with an high-end encoder. In performance applications, where low-speed performance & precision isn't of importance, sensorless control is a good option.

Voltage Level

Your power source & motor drive should have a high enough voltage rating to reach the required no-load max speed of your application. Use the equation below to calculate the required voltage level for your application.

$$VDC(V) = K_E(V \div kRPM) \times 1000 \times \text{Speed}(RPM)$$

Peak Current

The required peak current rating can be calculated with the equation below. Keep in mind that the equation is only valid for a ideal motor drive which means you should add a buffer of 15% to account for current ripple caused by switching.

$$\text{PeakCurrent}(A_{rms}) = \frac{\text{PeakTorque}(Nm)}{K_T(Nm \div A_{rms})}$$

Definitions & Glossary

Motor constant, K_m

The motor constant for the three-phase SlimTorq™ motors is given by the torque constant, & line to line resistance, see the equation below:

$$K_m (Nm \div \sqrt{W}) = \frac{K_T (Nm \div A_{rms})}{\sqrt{Line2LineResistance (Ohm) \times \frac{3}{2}}}$$

Note that the line to line resistance is converted to the three-phase equivalent of the resistance by multiplying it with the factor of 3/2.

Voltage constant, K_e

The voltage constant is based on peak value of the line-to-line voltage & the rotational speed.

$$K_E (V \div kRPM) = \frac{Line2LineVoltage (V)}{1000 \times RotationalSpeed (RPM)}$$

Torque constant, K_t

The torque constant is based on the torque output & RMS I_q current.

$$K_T (Nm \div A_{rms}) = \frac{Torque (Nm)}{I_q (A_{rms})}$$

Max continuous torque

The highest torque the motor can output continuously without overheating. The stated values assume that the motor is mounted in a small housing placed in an airflow with an airspeed of 5 m/s and an ambient temperature of 20°C.

Keep in mind that higher continuous torque can be expected if the motor is mounted in thermally conducting housing with larger cooling surface and increased airflow. The motors are not limited by saturation so the continuous torque is mostly dependant of the cooling ability of the overall assembly.

Peak torque

The peak torque assumes the motor starts from a rest with a motor temperature equivalent to the ambient temperature of 20°C.

Line to line resistance

The resistance measured at the motor terminals. Also called terminal resistance. The phase resistance can be calculated with the equations below.

Delta connections:

Phase resistance = Line-to-line resistance x 1.5

Star connections:

Phase resistance = Line-to-line resistance x 0.5

Electrical Time Constant

The electrical time constant (denoted as τ) of an electric motor or circuit is a measure of how quickly the current in the motor or circuit responds to changes in voltage. It indicates how fast the current can rise to its steady-state value after a voltage is applied.

In a motor, it's calculated as the ratio of the motor's inductance (L) to its resistance (R):

$$\tau = \frac{L}{R}$$

The time constant impacts how quickly a motor can accelerate or decelerate and how it will behave under dynamic load conditions.

Continuous power loss

The continuous power loss is the loss that is generated as heat at the rated continuous torque. The value is based on the assumption that the speed is equal to 0 RPM, i.e. speed dependant losses are not considered.

FAQs

Contact us today

For any additional inquiries not addressed on this page, please feel free to contact us at sales@alvaindustries.com. Our team will respond quickly to assist you.

How do I mechanically integrate the motor?

There are three general concepts for mechanical integration of the frameless motors.

1. Clamping
2. Adhesion
3. Interference fit

All three has their inherent advantages & considerations, please read more under [Mechanical Integration](#).

Do you offer custom motors?

Due the inherent simplicity & flexibility our manufacturing technology FiberPrinting™ we can offer fully custom solutions for your application. Please contact us at sales@alvaindustries.com to learn more.

Does the motors come with a temperature sensor?

Most of the SlimTorq™ motors can be ordered with a temperature sensor as an optional feature. Please see the [Ordering Alva Motors](#) section to see how you configure your product with a temperature sensor. The default temp sensor (A) is a 10K NTC with a B25/100 value of $3492 \pm 1\% K$ (TDK B57541G1103+000).

What is the airgap between rotor and stator?

The airgap dimension is dependant on the motor size, you can find the mechanical dimensions of the motor components in the CAD file.

What's required to drive the motors?

The SlimTorq™ motors are of the three-phase BLDC type & can be driven with most off-the-shelf controllers. The motors can be driven in trapezoidal, FOC, sinusoidal mode & with or without sensor feedback. Special considerations need to be taken regarding the switching frequency due to low inductance of the motors. Please read more under [Choosing a Motor Drive](#).

How do I select the correct motor for my application?

The easiest way to select the correct motor for your application is to use our [Product Selector](#). Insert your requirements in the Product Selector and it will find precisely the right motor for you. It will also allow you to fully configure the motor and order it. We are happy to help if you desire personal assistance to select the right motor, just contact us on sales@alvaindustries.com.

What's the typical delivery time for a standard motor?

Depending on the version of the motor, the lead time may vary. As a general rule of thumb, for units that are standard configurations, the lead time for small volumes (1-5 units) is around 8-10 weeks. For bigger volumes or small customizations the lead time may change, please reach out to us at sales@alvaindustries.com or call us.

FAQs

Contact us today

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Can I change the specifications of the motors?

Almost all the motor come in two main configurations: Lite and Max. Lite is the lightweight option & Max has been maximised in terms of motor constant and torque output.

Each motor has multiple winding configurations to align with your required max-speed. The motors can be configured in Star(Y) and Delta(D), and in different number of series-turns. The cables exit orientation also come in two variants: axial and radial.

The motors can be equipped with temperature sensors and different wire gauges.

To understand how to configure your product please see [Configured To Your Needs](#) section & to simplify the process check out our [Product Selector](#).

Please contact us at sales@alvaindustries.com if there isn't a configuration that meets your needs. Our FiberPrinting™ technology allows us to quickly develop custom motors according to your requirements.

What are the tolerances for the mounting?

You can find the mechanical tolerances of the motor components in the mechanical section of each product page. For more information how to mechanically integrate the motors into your application please see [Mechanical Integration](#) section. The STEP files of the motors are available on the [SlimTorq™ product page](#).

Are these motors more expensive than other slotless motors?

No. Alva's manufacturing methods, lead by the FiberPrinting™ technology, are comparable in cost & price with motors with similar characteristics (thin, light, cogging free slotless frameless motors).

Are slotless motors more expensive than slotted motors?

Alva's FiberPrinting™ manufacturing technology for slotless stators & proprietary manufacturing method of Halbach array rotors, allow for production cost that can be quite competitive to even manufacturing methods of slotted motors, enabling also competitive pricing. On the other hand, slotless motors from other manufacturers tend to be more expensive than slotted motors due to the larger magnets & legacy manufacturing methods.

Are these motors available for high voltage?

All the SlimTorq™ motors are rated with a maximum voltage level of 72V. Higher voltage ratings will be available in the future. Please contact sales@alvaindustries.com if your application require a higher voltage rating.

FAQs

Contact us today

For any additional inquiries not addressed on this page, please feel free to contact us at sales@alvaindustries.com. Our team will respond quickly to assist you.

Do I need to have active cooling on the motors to achieve the specifications of Torque/speed?

The continuous torque and speed values in the product guide are based on the assumption that motors are mounted in a small housing and with some airflow passing over the housing. To understand how the motor will perform in your application please insert your requirements and select the correct cooling conditions in our [Product Selector](#). Or contact us at sales@alvaindustries.com & we can support you with a detailed thermal & performance study.

Can I use the motors with magnetic encoders?

Yes, the SlimTorq™ motors can be combined with magnetic encoders. It's important to review the requirements of the magnetic encoder in terms of maximum external magnetic field. The table below shows minimum required distance measured from the stator edge to the magnetic encoder.

25-9-M	1.8 mm	105-17-L	2.6 mm
39-10-M	1.75 mm	105-17-M	3.6 mm
51-12-M	1.9 mm	130-27-L	0 mm
75-20-L	2 mm	130-27-M	2.2 mm
75-20-M	3.4 mm	160-17-L	2.9 mm
85-24-L	3.2 mm	160-17-M	4.3 mm
85-24-M	4.3 mm		

Are there possibilities for Custom Mounting solutions?

Our expertise not only lies on the design and manufacture of motors but also in their integration into diverse types of housings and mechatronics designs. Given that the possibilities to mount a frameless motor may vary, we give standard recommendations on how customers can do that on their own: [Mechanical Integration](#). However, we are also capable of modifying the mounting possibilities of the motors, according to the applications needs. Reach out to: sales@alvaindustries.com or call us.

Can I combine these motors with a gearbox?

With a large inner-diameter and low-speed dependant losses, the SlimTorq™ motors are highly suitable to be combined with gearboxes for torque demanding applications. They can be combined with all the typical type of gearboxes for precision applications such as planetary-, worm-, strain wave-, bevel- and cycloidal-gearbox.

What is the life time of frameless motors?

The SlimTorq™ motors are brushless motors, they are therefore inherently not exposed to any mechanical wear assuming a well designed mechanical assembly. The Litz Wire of the stator windings are isolated with of a P180 enamel according to IEC 60172, with a rated lifetime of 20,000 hrs at a continuous temperature of 180 °C.

As long as the motors doesn't exceed their maximum rated temperature of 140°C, the motors will have next to unlimited life-time.



Contact

Headquarters

Fossegrenda 1E, 7038 Trondheim, Norway
+47 73 204 501
sales@alvaindustries.com
www.alvaindustries.com

US Office

611 Gateway Blvd, Suite 120, South San Francisco,
California, 94080, US
+1 (650) 622 5719
sales@alvaindustries.com

